

A Tensor Formulation of the Choi–Jamiołkowski Isomorphism for Quantum and Hybrid Networks

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We develop a unified operator and tensor framework for the local factors of classical, quantum, and hybrid Bayesian networks. Finite probability distributions are represented as diagonal density operators, while conditional probability distributions give rise to positive conditional operators that, through the Choi–Jamiołkowski isomorphism, coincide with the Choi operators of the quantum channels induced by the corresponding stochastic maps. This identifies classical conditional probabilities as a special class of quantum channels and places classical and quantum nodes of a directed acyclic graph within a common positive-operator formalism. Representing these operators as tensors in fixed orthonormal bases, we show that channel action reduces to a contraction over the input indices, providing a unified computational language for network composition. Classical Bayesian-network factor multiplication, quantum channel composition, and the interaction of state preparation and measurement in hybrid networks all emerge as instances of the same tensor-contraction rules.

I. INTRODUCTION

Bayesian networks describe probabilistic systems by means of directed acyclic graphs, with a local probability distribution associated with each node, conditioned on its parents [1, Sec. 11.2.3]. When quantum systems are included in the network, their local transformations are described by quantum channels rather than by conditional probability distributions. The starting point of this work is to ask whether these classical and quantum factors can be represented in a common operator language and combined within the same component formalism.

A finite probability distribution can be represented by a density operator that is diagonal in the basis indexed by its possible values. Applying the same construction to a conditional distribution $p_{X|Y}$ gives a positive operator $\rho_{X|Y}$ satisfying

$$\mathrm{Tr}_X (\rho_{X|Y}) = \mathrm{Id}_Y. \quad (1)$$

This conditional operator is not merely another way of recording the conditional probabilities. Its positivity and normalization are precisely the conditions required of the Choi operator of a quantum channel. Through the Choi–Jamiołkowski isomorphism, $\rho_{X|Y}$ is the Choi operator of the channel induced by the stochastic map associated with $p_{X|Y}$. The stochastic maps defined by classical conditional distributions therefore embed into quantum channels.

This observation gives a common representation of the local factors associated with the nodes of a directed acyclic graph. A root factor is represented by a density operator, a classical conditional factor by a diagonal Choi operator, and a quantum conditional factor by the Choi operator of a quantum channel. All these local factors are positive operators, while the distinction between the classical and quantum cases is expressed by their structure

and normalization. This point of view is related to the conditional-state formalism of Leifer and Spekkens [2] and to the Q-factor framework of Di Guardia, Ehrhard, and Faggian [3].

Representing all local factors by operators leaves the question of how to combine them. If $J(\Phi)$ is the Choi operator of a channel $\Phi : L(\mathcal{Y}) \rightarrow L(\mathcal{X})$, its action is

$$\Phi(A) = \mathrm{Tr}_Y (J(\Phi) (\mathrm{Id}_X \otimes A^T)) \quad (2)$$

[4, Eq. (2.66)]. Thus, the input is transposed, lifted to the space of the Choi operator, multiplied by that operator, and finally removed by a partial trace. For a single channel these operations are direct, but when several channels are composed the registers involved in each step become less apparent.

We therefore write the operators as tensors in fixed orthonormal bases. The indices identify the registers to which the components belong, and the action of a Choi operator is expressed directly by contracting its input indices. The same description applies to channel composition: the indices associated with an intermediate quantum register are contracted between the corresponding channel factors. The same component notation can be used for classical and hybrid networks, although the operations in the two cases are not identical. For diagonal operators, factor multiplication reduces to the pointwise product used in classical Bayesian networks. In a hybrid network, classically controlled state preparations, quantum channel factors, and destructive measurements are combined by contracting the intermediate quantum indices. The emphasis of this paper is therefore on the representation and composition of local classical and quantum factors, rather than on Bayesian inference.

Sections II–IV introduce the notation and mathematical background used in the paper. Section V establishes the connection between conditional distributions, conditional operators, and the quantum channels induced by stochastic maps. Section VI develops the tensor formulation for classical Bayesian networks, quantum channel networks, and hybrid classical–quantum networks.

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II. PRELIMINARIES AND NOTATION

All vector spaces considered below are finite-dimensional complex Euclidean spaces. We follow the notation and conventions of Watrous [4, Secs. 1.1.1–1.1.2 and 1.2.1], with minor adaptations for consistency with the tensor notation used in this paper. This section fixes the notation needed for linear operators, maps on operator spaces, and probability vectors.

A. Complex Euclidean spaces

An *alphabet* is a finite, nonempty set whose elements are called symbols. Alphabets are denoted by capital Greek letters such as Σ and Γ . For an alphabet Σ , define

$$\mathbb{C}^\Sigma = \{u : \Sigma \rightarrow \mathbb{C}\}. \quad (3)$$

Equipped with pointwise addition and scalar multiplication and with the standard inner product, $\mathbb{C}^\Sigma$ is a complex Euclidean space of dimension $|\Sigma|$. Complex Euclidean spaces are denoted by script letters such as $\mathcal{X}$, $\mathcal{Y}$, and $\mathcal{Z}$. We denote the standard orthonormal basis of $\mathbb{C}^\Sigma$ by $\{e_a : a \in \Sigma\}$.

B. Linear operators

For complex Euclidean spaces $\mathcal{X}$ and $\mathcal{Y}$, the space of linear operators from $\mathcal{X}$ to $\mathcal{Y}$ is denoted

$$L(\mathcal{X}, \mathcal{Y}) = \{A : \mathcal{X} \rightarrow \mathcal{Y} : A \text{ is linear}\}. \quad (4)$$

We write $L(\mathcal{X}) = L(\mathcal{X}, \mathcal{X})$, and denote the identity operator on $\mathcal{X}$ by $\text{Id}_{\mathcal{X}}$. Let $\mathcal{X} = \mathbb{C}^\Sigma$ and $\mathcal{Y} = \mathbb{C}^\Gamma$. For $a \in \Gamma$ and $b \in \Sigma$, the matrix unit $E_{a,b} \in L(\mathcal{X}, \mathcal{Y})$ is defined by

$$E_{a,b}(c, d) = \begin{cases} 1, & (c, d) = (a, b), \\ 0, & \text{otherwise.} \end{cases} \quad c \in \Gamma, \quad d \in \Sigma. \quad (5)$$

The collection $\{E_{a,b} : a \in \Gamma, b \in \Sigma\}$ is the standard basis of $L(\mathcal{X}, \mathcal{Y})$.

For $A \in L(\mathcal{X}, \mathcal{Y})$, the transpose relative to the standard bases is the operator $A^T \in L(\mathcal{Y}, \mathcal{X})$ defined by

$$A^T(b, a) = A(a, b), \quad a \in \Gamma, \quad b \in \Sigma. \quad (6)$$

An operator $H \in L(\mathcal{X})$ is *Hermitian* when $H = H^\dagger$. The trace of $X \in L(\mathcal{X})$ is

$$\text{Tr}(X) = \sum_{a \in \Sigma} X(a, a), \quad \mathcal{X} = \mathbb{C}^\Sigma. \quad (7)$$

1. Positive, density, and diagonal operators

1. *Positive semidefinite operators.* An operator $P \in L(\mathcal{X})$ is positive semidefinite if $P = A^\dagger A$ for some

$A \in L(\mathcal{X})$. We denote the cone of positive semidefinite operators by

$$\text{Pos}(\mathcal{X}) = \{A^\dagger A : A \in L(\mathcal{X})\}. \quad (8)$$

Equivalently, P is positive semidefinite if and only if it is Hermitian and all its eigenvalues are nonnegative.

2. *Density operators.* A density operator is a positive semidefinite operator of trace one. The set of density operators on $\mathcal{X}$ is

$$D(\mathcal{X}) = \{\rho \in \text{Pos}(\mathcal{X}) : \text{Tr}(\rho) = 1\}. \quad (9)$$

3. *Diagonal operators.* An operator $X \in L(\mathbb{C}^\Sigma)$ is diagonal if $X(a, b) = 0$ whenever $a \neq b$.

C. Linear maps on operator spaces

For complex Euclidean spaces $\mathcal{X}$ and $\mathcal{Y}$, let

$$T(\mathcal{X}, \mathcal{Y}) = L(L(\mathcal{X}), L(\mathcal{Y})) \quad (10)$$

denote the vector space of linear maps $\Phi : L(\mathcal{X}) \rightarrow L(\mathcal{Y})$. We denote the identity map on $L(\mathcal{X})$ by $\text{Id}_{L(\mathcal{X})}$. Upon identifying $L(\mathbb{C})$ with $\mathbb{C}$, the trace is an element of

$$\text{Tr} \in T(\mathcal{X}, \mathbb{C}). \quad (11)$$

A map $\Phi \in T(\mathcal{X}, \mathcal{Y})$ is *positive* if

$$\Phi(P) \in \text{Pos}(\mathcal{Y}) \quad \text{for every } P \in \text{Pos}(\mathcal{X}). \quad (12)$$

It is *trace-preserving* if

$$\text{Tr}(\Phi(X)) = \text{Tr}(X) \quad \text{for every } X \in L(\mathcal{X}). \quad (13)$$

D. Probability vectors

For an alphabet Σ , a vector $p \in \mathbb{R}^\Sigma$ is a *probability vector* if

$$p(a) \geq 0 \quad \text{for every } a \in \Sigma, \quad \sum_{a \in \Sigma} p(a) = 1. \quad (14)$$

The set of all probability vectors on Σ is denoted $\mathcal{P}(\Sigma)$.

III. TENSOR PRODUCTS AND CONTRACTIONS

Composite quantum systems, Choi operators, and the tensor contractions used later all rely on the same finite-dimensional tensor-product structure. We therefore fix the relevant conventions here, including the relation between operators and tensors. The conventions for tensor products and maps on operator spaces follow Watrous [4, Secs. 1.1.1–1.1.2].

A. Tensor-product conventions

Let n be a positive integer and let $\mathcal{X}_k = \mathbb{C}^{\Sigma_k}$ for $k \in \{1, \dots, n\}$. We identify

$$\mathcal{X}_1 \otimes \dots \otimes \mathcal{X}_n = \mathbb{C}^{\Sigma_1 \times \dots \times \Sigma_n}. \quad (15)$$

For vectors $u_k \in \mathcal{X}_k$, we write $u_1 \otimes \dots \otimes u_n$ for their elementary tensor.

For $A \in L(\mathcal{X}, \mathcal{Y})$ and $B \in L(\mathcal{Z}, \mathcal{W})$, their tensor product is the operator

$$A \otimes B \in L(\mathcal{X} \otimes \mathcal{Z}, \mathcal{Y} \otimes \mathcal{W}) \quad (16)$$

characterized by

$$(A \otimes B)(u \otimes v) = Au \otimes Bv. \quad (17)$$

The same notation is used for tensor products of more than two operators.

For $\Phi \in T(\mathcal{X}, \mathcal{Y})$ and $\Psi \in T(\mathcal{Z}, \mathcal{W})$, their tensor product is the unique linear map

$$\Phi \otimes \Psi \in T(\mathcal{X} \otimes \mathcal{Z}, \mathcal{Y} \otimes \mathcal{W}) \quad (18)$$

such that

$$(\Phi \otimes \Psi)(X \otimes Z) = \Phi(X) \otimes \Psi(Z) \quad (19)$$

for every $X \in L(\mathcal{X})$ and $Z \in L(\mathcal{Z})$.

1. Completely positive maps

A map $\Phi \in T(\mathcal{X}, \mathcal{Y})$ is *completely positive* if

$$\Phi \otimes \text{Id}_{L(\mathcal{Z})} \in T(\mathcal{X} \otimes \mathcal{Z}, \mathcal{Y} \otimes \mathcal{Z}) \quad (20)$$

is positive for every complex Euclidean space $\mathcal{Z}$. The set of all such maps is denoted $\text{CP}(\mathcal{X}, \mathcal{Y})$.

2. Partial trace

For complex Euclidean spaces $\mathcal{X}$ and $\mathcal{Y}$, the partial trace over $\mathcal{X}$ is

$$\text{Tr}_{\mathcal{X}} := \text{Tr} \otimes \text{Id}_{L(\mathcal{Y})} \in T(\mathcal{X} \otimes \mathcal{Y}, \mathcal{Y}). \quad (21)$$

It is characterized on elementary operators by

$$\text{Tr}_{\mathcal{X}}(A \otimes B) = \text{Tr}(A)B, \quad A \in L(\mathcal{X}), \quad B \in L(\mathcal{Y}). \quad (22)$$

Similarly,

$$\text{Tr}_{\mathcal{Y}} := \text{Id}_{L(\mathcal{X})} \otimes \text{Tr} \in T(\mathcal{X} \otimes \mathcal{Y}, \mathcal{X}). \quad (23)$$

B. Tensors and contractions

The tensor notation used later keeps track of the input and output indices associated with each subsystem. We only recall the conventions needed below; more detailed treatments are given by Greub [5, Secs. 3.12–3.13], Wendl [6, Secs. A.6–A.7, pp. 175–180], and Dullemond and Peeters [7, pp. 13 and 21–22].

Definition III.1. Let $V_1, \dots, V_m$ and $W_1, \dots, W_n$ be finite-dimensional complex vector spaces. A tensor of type (m, n) is an element

$$T \in (V_1 \otimes \dots \otimes V_m) \otimes (W_1^* \otimes \dots \otimes W_n^*). \quad (24)$$

Its order is $m + n$. Given bases $\{e_{i_a}^{(a)}\}$ of V_a and $\{f_{j_b}^{(b)}\}$ of W_b , with dual bases $\{f_{(b)}^{j_b}\}$, the components of T are defined by

$$T = \sum_{\substack{i_1, \dots, i_m \\ j_1, \dots, j_n}} T^{i_1 \dots i_m}_{j_1 \dots j_n} (e_{i_1}^{(1)} \otimes \dots \otimes e_{i_m}^{(m)}) \otimes (f_{(1)}^{j_1} \otimes \dots \otimes f_{(n)}^{j_n}). \quad (25)$$

The upper indices are contravariant and label the vector factors, whereas the lower indices are covariant and label the dual-vector factors.

A contraction may pair only a contravariant index with a covariant index referring to the same vector space [7, p. 22]. For example, if $V_m = W_1$ and the corresponding bases agree, then

$$S^{i_1 \dots i_{m-1}}_{j_2 \dots j_n} = T^{i_1 \dots i_{m-1} k}_{k j_2 \dots j_n}. \quad (26)$$

The repeated index k is summed over, while the remaining indices are free. This is the Einstein summation convention, which we use unless a different convention is stated explicitly.

For finite-dimensional complex vector spaces V and W , there is a canonical, basis-independent isomorphism [8, Sec. 1]

$$L(V, W) \simeq W \otimes V^* \quad (27)$$

under which every linear operator corresponds to a unique tensor, and conversely [5, Sec. 1.28]. This identification allows us to regard an operator on $\mathcal{X}_1 \otimes \dots \otimes \mathcal{X}_r$ as a tensor carrying one upper and one lower index for each subsystem, as we will exploit later.

IV. QUANTUM STATES, CHANNELS, AND MEASUREMENTS

We associate each register with a finite-dimensional complex Euclidean space. Quantum states of the register are represented by density operators, while transformations between registers are represented by quantum channels. We also introduce the description of measurement outcomes that will be needed for the classical–quantum networks considered later. Our conventions follow Watrous [4, Secs. 2.1.1–2.1.2].

A. Registers and states

1. Registers and classical state sets

Definition IV.1. Registers are defined as follows:

1. every alphabet Σ is a register;
2. if n is a positive integer and $Y_1, \dots, Y_n$ are registers, then $X = (Y_1, \dots, Y_n)$ is a register.

Registers of the first type are called *simple*, whereas tuples of registers are called *compound*.

Definition IV.2. The *classical state set* of a register X is specified recursively:

1. if $X = \Sigma$ is simple, its classical state set is Σ ;
2. if $X = (Y_1, \dots, Y_n)$ is compound and Σ_k is the classical state set of Y_k , then the classical state set of X is

$$\Sigma_1 \times \dots \times \Sigma_n. \quad (28)$$

The elements of this set are the classical states of the register. In particular, a classical state of a compound register specifies one classical state for each of its constituent registers.

2. Quantum states of registers

Definition IV.3. If Σ is the classical state set of a register X , the complex Euclidean space associated with X is

$$\mathcal{X} = \mathbb{C}^\Sigma. \quad (29)$$

Thus, if $X = (Y_1, \dots, Y_n)$ and $\mathcal{Y}_k = \mathbb{C}^{\Sigma_k}$ is the space associated with Y_k , the tensor-product convention fixed above gives

$$\mathcal{X} = \mathbb{C}^{\Sigma_1 \times \dots \times \Sigma_n} = \mathcal{Y}_1 \otimes \dots \otimes \mathcal{Y}_n. \quad (30)$$

Definition IV.4. A *quantum state* of a register X is a density operator $\rho \in D(\mathcal{X})$, where $\mathcal{X}$ is the complex Euclidean space associated with X .

a. Mixtures. The set $D(\mathcal{X})$ is convex. Thus, let Γ be an alphabet, let $\{\rho_a : a \in \Gamma\} \subseteq D(\mathcal{X})$, and let $p \in \mathcal{P}(\Gamma)$. The operator

$$\rho = \sum_{a \in \Gamma} p(a) \rho_a \quad (31)$$

belongs to $D(\mathcal{X})$ and is called the mixture of the states ρ_a with probabilities $p(a)$.

B. Quantum channels

Quantum channels model transformations of register states. We use the standard completely-positive, trace-preserving definition [4, Def. 2.13].

Definition IV.5. A *quantum channel* from $\mathcal{X}$ to $\mathcal{Y}$ is a linear map

$$\Phi : L(\mathcal{X}) \rightarrow L(\mathcal{Y}) \quad (32)$$

that is completely positive and trace-preserving; equivalently, $\Phi \in \text{CP}(\mathcal{X}, \mathcal{Y})$ and

$$\text{Tr}(\Phi(X)) = \text{Tr}(X) \quad \text{for every } X \in L(\mathcal{X}). \quad (33)$$

A channel maps density operators to density operators. Consequently, if X and Y have associated spaces $\mathcal{X}$ and $\mathcal{Y}$, respectively, and X is prepared in the state $\rho \in D(\mathcal{X})$, then the output state is $\Phi(\rho) \in D(\mathcal{Y})$.

C. Measurements

The outcome statistics of a quantum measurement are described by a positive operator-valued measure (POVM) [4, Def. 2.34]. In this paper, measurements are used as destructive quantum-to-classical processes: only the classical outcome is retained, while the postmeasurement quantum system is discarded.

Definition IV.6. A *POVM* on $\mathcal{X}$ with outcome alphabet Σ is a function

$$\mu : \Sigma \rightarrow \text{Pos}(\mathcal{X}) \quad (34)$$

satisfying

$$\sum_{a \in \Sigma} \mu(a) = \text{Id}_{\mathcal{X}}. \quad (35)$$

If the measured state is $\rho \in D(\mathcal{X})$, the outcome distribution $p \in \mathcal{P}(\Sigma)$ is given by

$$p(a) = \text{Tr}(\mu(a)\rho). \quad (36)$$

The operator $\mu(a)$ is the POVM element associated with outcome a . We may equivalently specify the measurement as a collection $\{P_a : a \in \Sigma\} \subseteq \text{Pos}(\mathcal{X})$, where $P_a = \mu(a)$ and $\sum_{a \in \Sigma} P_a = \text{Id}_{\mathcal{X}}$.

V. CLASSICAL PROBABILITY IN OPERATOR FORM

Finite probability distributions can be represented by diagonal density operators. We apply this representation to joint and conditional distributions. The resulting conditional operator will then be identified, through the Choi–Jamiołkowski isomorphism, with the Choi operator of the quantum channel induced by the corresponding stochastic map.

A. Finite probability distributions

Let $\mathbf{X}$ be a register with classical state set Σ . A probabilistic state of $\mathbf{X}$ is a probability vector $p_{\mathbf{X}} \in \mathcal{P}(\Sigma)$, where $p_{\mathbf{X}}(x)$ is the probability that $\mathbf{X}$ is in the classical state $x \in \Sigma$.

a. Joint and marginal distributions. Let $\mathbf{X}$ and $\mathbf{Y}$ be registers with classical state sets Σ and Γ , respectively. The compound register $(\mathbf{X}, \mathbf{Y})$ has classical state set $\Sigma \times \Gamma$, so a joint probabilistic state is a probability vector

$$p_{\mathbf{X}, \mathbf{Y}} \in \mathcal{P}(\Sigma \times \Gamma). \quad (37)$$

The value $p_{\mathbf{X}, \mathbf{Y}}(x, y)$ is the probability associated with the joint classical state (x, y) , and therefore

$$\sum_{x \in \Sigma} \sum_{y \in \Gamma} p_{\mathbf{X}, \mathbf{Y}}(x, y) = 1. \quad (38)$$

The corresponding marginal probabilistic states are

$$p_{\mathbf{X}}(x) = \sum_{y \in \Gamma} p_{\mathbf{X}, \mathbf{Y}}(x, y), \quad (39)$$

for every $x \in \Sigma$, and

$$p_{\mathbf{Y}}(y) = \sum_{x \in \Sigma} p_{\mathbf{X}, \mathbf{Y}}(x, y), \quad (40)$$

for every $y \in \Gamma$.

b. Conditional distributions. A conditional probabilistic state can be viewed as a function of the form

$$p_{\mathbf{X}|\mathbf{Y}} : \Gamma \rightarrow \mathcal{P}(\Sigma). \quad (41)$$

In other words, for each fixed $y \in \Gamma$, this function assigns a probability state $p_{\mathbf{X}|\mathbf{Y}}(\cdot | y) \in \mathcal{P}(\Sigma)$ to $\mathbf{X}$. Whenever $p_{\mathbf{Y}}(y) > 0$, its components are determined by the joint distribution through

$$p_{\mathbf{X}|\mathbf{Y}}(x|y) = \frac{p_{\mathbf{X}, \mathbf{Y}}(x, y)}{p_{\mathbf{Y}}(y)}. \quad (42)$$

Since $p_{\mathbf{X}|\mathbf{Y}}(\cdot | y)$ is a probability state, we must have

$$\sum_{x \in \Sigma} p_{\mathbf{X}|\mathbf{Y}}(x|y) = 1 \quad \text{for every } y \in \Gamma. \quad (43)$$

If $p_{\mathbf{Y}}(y) = 0$, the quotient above is not defined and the joint distribution does not determine the conditional state at y . In that case, any probability vector in $\mathcal{P}(\Sigma)$ may be assigned to $p_{\mathbf{X}|\mathbf{Y}}(\cdot | y)$.

B. Probability distributions as diagonal operators

a. Single-register distributions. Let $\mathcal{X} = \mathbb{C}^{\Sigma}$ be the complex Euclidean space associated with $\mathbf{X}$. Relative to its standard basis, the classical state $a \in \Sigma$ is represented

by the density operator $E_{a,a}$. Therefore, the probability vector $p_{\mathbf{X}}$ is represented by

$$\rho_{\mathbf{X}} = \sum_{a \in \Sigma} p_{\mathbf{X}}(a) E_{a,a}. \quad (44)$$

Indeed, $\rho_{\mathbf{X}}$ is positive semidefinite because it is a nonnegative linear combination of the operators $E_{a,a}$, and

$$\text{Tr}(\rho_{\mathbf{X}}) = \sum_{a \in \Sigma} p_{\mathbf{X}}(a) = 1. \quad (45)$$

Thus $\rho_{\mathbf{X}} \in D(\mathcal{X})$.

b. Joint distributions and marginalization. Let $\mathbf{Y}$ have classical state set Γ and associated space $\mathcal{Y} = \mathbb{C}^{\Gamma}$. The compound register $(\mathbf{X}, \mathbf{Y})$ is associated with

$$\mathbb{C}^{\Sigma \times \Gamma} = \mathcal{X} \otimes \mathcal{Y}. \quad (46)$$

Each joint classical state (x, y) is represented by $E_{x,x} \otimes E_{y,y}$. Therefore the joint distribution $p_{\mathbf{X}, \mathbf{Y}}$ is represented by the quantum state

$$\rho_{\mathbf{X}, \mathbf{Y}} = \sum_{x \in \Sigma} \sum_{y \in \Gamma} p_{\mathbf{X}, \mathbf{Y}}(x, y) (E_{x,x} \otimes E_{y,y}). \quad (47)$$

This operator is positive semidefinite, and its trace is

$$\begin{aligned} \text{Tr}(\rho_{\mathbf{X}, \mathbf{Y}}) &= \sum_{x \in \Sigma} \sum_{y \in \Gamma} p_{\mathbf{X}, \mathbf{Y}}(x, y) \text{Tr}(E_{x,x} \otimes E_{y,y}) \\ &= \sum_{x \in \Sigma} \sum_{y \in \Gamma} p_{\mathbf{X}, \mathbf{Y}}(x, y) = 1. \end{aligned} \quad (48)$$

Hence $\rho_{\mathbf{X}, \mathbf{Y}} \in D(\mathcal{X} \otimes \mathcal{Y})$.

How can we recover a marginal distribution from this joint quantum state? In the classical setting, marginalization over $\mathbf{Y}$ gives

$$\sum_{y \in \Gamma} p_{\mathbf{X}, \mathbf{Y}}(x, y) = p_{\mathbf{X}}(x). \quad (49)$$

One may see that marginalization can be regarded as the map

$$\Phi : \mathcal{P}(\Sigma \times \Gamma) \rightarrow \mathcal{P}(\Sigma). \quad (50)$$

This map is affine on the set of probability vectors. In our quantum setting, we therefore need a corresponding map

$$\Psi : D(\mathcal{X} \otimes \mathcal{Y}) \rightarrow D(\mathcal{X}). \quad (51)$$

Such a map exists and it is exactly the partial trace:

$$\Psi = \text{Tr}_{\mathcal{Y}} = \text{Id}_{L(\mathcal{X})} \otimes \text{Tr}. \quad (52)$$

More precisely, Ψ is the restriction to density operators of the linear partial-trace map defined on $L(\mathcal{X} \otimes \mathcal{Y})$. Thus,

we have

$$\text{Tr}_{\mathcal{Y}}(\rho_{\mathbf{X},\mathbf{Y}}) = \sum_{x \in \Sigma} \sum_{y \in \Gamma} p_{\mathbf{X},\mathbf{Y}}(x, y) \text{Tr}(E_{y,y}) E_{x,x} \quad (53)$$

$$= \sum_{x \in \Sigma} \left(\sum_{y \in \Gamma} p_{\mathbf{X},\mathbf{Y}}(x, y) \right) E_{x,x} \quad (54)$$

$$= \sum_{x \in \Sigma} p_{\mathbf{X}}(x) E_{x,x} = \rho_{\mathbf{X}}. \quad (55)$$

The quantum analogue of marginalization is therefore

$$\rho_{\mathbf{X}} = \text{Tr}_{\mathcal{Y}}(\rho_{\mathbf{X},\mathbf{Y}}). \quad (56)$$

The resulting state is called the reduction of $\rho_{\mathbf{X},\mathbf{Y}}$ to the register $\mathbf{X}$ [4, Sec. 2.1.3].

1. Families of conditional states

We can now apply the same diagonal representation to every probability vector in the conditional family. For each fixed $y \in \Gamma$, define

$$\rho_{\mathbf{X}|y} = \sum_{x \in \Sigma} p_{\mathbf{X}|\mathbf{Y}}(x|y) E_{x,x}. \quad (57)$$

Because $p_{\mathbf{X}|\mathbf{Y}}(\cdot|y)$ is a probability vector, $\rho_{\mathbf{X}|y} \in D(\mathcal{X})$ for every $y \in \Gamma$. At this stage, the value of $\mathbf{Y}$ is fixed, and the conditional distribution is represented by the family of density operators

$$\{\rho_{\mathbf{X}|y} : y \in \Gamma\} \subseteq D(\mathcal{X}). \quad (58)$$

2. Convex mixtures of conditional states

What happens if $\mathbf{Y}$ is itself distributed according to the probability vector $p_{\mathbf{Y}} \in \mathcal{P}(\Gamma)$? The corresponding mixture of conditional quantum states is

$$\rho = \sum_{y \in \Gamma} p_{\mathbf{Y}}(y) \rho_{\mathbf{X}|y}. \quad (59)$$

This operator belongs to $D(\mathcal{X})$. Substituting Eq. (57) into Eq. (59), and using

$$p_{\mathbf{X},\mathbf{Y}}(x, y) = p_{\mathbf{Y}}(y) p_{\mathbf{X}|\mathbf{Y}}(x|y), \quad (60)$$

gives

$$\rho = \sum_{y \in \Gamma} p_{\mathbf{Y}}(y) \left(\sum_{x \in \Sigma} p_{\mathbf{X}|\mathbf{Y}}(x|y) E_{x,x} \right) \quad (61)$$

$$= \sum_{y \in \Gamma} \sum_{x \in \Sigma} p_{\mathbf{Y}}(y) p_{\mathbf{X}|\mathbf{Y}}(x|y) E_{x,x} \quad (62)$$

$$= \sum_{x \in \Sigma} \left(\sum_{y \in \Gamma} p_{\mathbf{X},\mathbf{Y}}(x, y) \right) E_{x,x} \quad (63)$$

$$= \sum_{x \in \Sigma} p_{\mathbf{X}}(x) E_{x,x} \quad (64)$$

$$= \rho_{\mathbf{X}}. \quad (65)$$

This is the operator form of the law of total probability.

3. Ensembles of conditional states

The weighted family in Eq. (59) may equivalently be described by the ensemble

$$\eta : \Gamma \rightarrow \text{Pos}(\mathcal{X}), \quad \eta(y) = p_{\mathbf{Y}}(y) \rho_{\mathbf{X}|y}. \quad (66)$$

It satisfies the normalization condition

$$\text{Tr} \left(\sum_{y \in \Gamma} \eta(y) \right) = 1, \quad (67)$$

because

$$\begin{aligned} \text{Tr} \left(\sum_{y \in \Gamma} \eta(y) \right) &= \sum_{y \in \Gamma} p_{\mathbf{Y}}(y) \text{Tr}(\rho_{\mathbf{X}|y}) \\ &= \sum_{y \in \Gamma} p_{\mathbf{Y}}(y) = 1. \end{aligned} \quad (68)$$

The trace $\text{Tr}(\eta(y)) = p_{\mathbf{Y}}(y)$ is the probability of the label y . Whenever $p_{\mathbf{Y}}(y) > 0$, the corresponding normalized conditional state is recovered through

$$\frac{\eta(y)}{\text{Tr}(\eta(y))} = \rho_{\mathbf{X}|y}. \quad (69)$$

If $p_{\mathbf{Y}}(y) = 0$, then $\eta(y) = 0$, and no normalized state can be recovered from that ensemble element; the conditional state at this zero-probability value remains the arbitrary choice made above.

4. The conditional operator

There is a second, equivalent way to represent the same conditional probabilities. Instead of viewing $p_{\mathbf{X}|\mathbf{Y}}$ as a map from Γ to $\mathcal{P}(\Sigma)$, we may view its components as the pointwise function

$$p_{\mathbf{X}|\mathbf{Y}} : \Sigma \times \Gamma \rightarrow [0, 1]. \quad (70)$$

The resulting array can be encoded as the operator

$$\rho_{\mathbf{X}|\mathbf{Y}} = \sum_{x \in \Sigma} \sum_{y \in \Gamma} p_{\mathbf{X}|\mathbf{Y}}(x|y) (E_{x,x} \otimes E_{y,y}) \quad (71)$$

in $L(\mathcal{X} \otimes \mathcal{Y})$. In terms of the family defined in Eq. (57), this operator has the simpler form

$$\rho_{\mathbf{X}|\mathbf{Y}} = \sum_{y \in \Gamma} \rho_{\mathbf{X}|y} \otimes E_{y,y}. \quad (72)$$

It is positive semidefinite because every coefficient $p_{\mathbf{X}|\mathbf{Y}}(x|y)$ is nonnegative. Its normalization is expressed

by a partial trace rather than by a unit trace.

$$\mathrm{Tr}_{\mathcal{X}}(\rho_{\mathcal{X}|\mathcal{Y}}) = \sum_{y \in \Gamma} \mathrm{Tr}(\rho_{\mathcal{X}|y}) E_{y,y} \quad (73)$$

$$= \sum_{y \in \Gamma} E_{y,y} \quad (74)$$

$$= \mathrm{Id}_{\mathcal{Y}}. \quad (75)$$

Consequently,

$$\mathrm{Tr}(\rho_{\mathcal{X}|\mathcal{Y}}) = \dim(\mathcal{Y}) = |\Gamma|. \quad (76)$$

Thus $\rho_{\mathcal{X}|\mathcal{Y}}$ is not a density operator. We call it the *conditional operator* of $\mathcal{X}$ given $\mathcal{Y}$.

The operator (71) therefore contains the same information as the family $\{\rho_{\mathcal{X}|y} : y \in \Gamma\}$ from Section VB 1, but represents all the conditional distributions simultaneously.

At the classical level, $p_{\mathcal{X}|\mathcal{Y}}$ also determines a stochastic map that sends an input distribution $p_{\mathcal{Y}}$ to the corresponding output distribution $p_{\mathcal{X}}$ [9, Sec. “Probability Theory”]. Can $\rho_{\mathcal{X}|\mathcal{Y}}$ similarly determine a map from $L(\mathcal{Y})$ to $L(\mathcal{X})$? This is precisely the role of the Choi–Jamiołkowski isomorphism. As we show next, the positivity and normalization of $\rho_{\mathcal{X}|\mathcal{Y}}$ imply that it is the Choi operator of a quantum channel.

C. Stochastic maps as quantum channels

1. The Choi–Jamiołkowski isomorphism

Let $\mathcal{X} = \mathbb{C}^{\Sigma}$ and $\mathcal{Y} = \mathbb{C}^{\Gamma}$. The map–operator correspondence used below originates in the work of Jamiołkowski and Choi [10, 11]. With the convention adopted here, the Choi–Jamiołkowski isomorphism is the linear bijection

$$J : T(\mathcal{Y}, \mathcal{X}) \rightarrow L(\mathcal{X} \otimes \mathcal{Y}) \quad (77)$$

defined by

$$J(\Phi) = \sum_{a \in \Gamma} \sum_{b \in \Gamma} \Phi(E_{a,b}) \otimes E_{a,b}. \quad (78)$$

The operator $J(\Phi)$ is called the *Choi operator* of Φ . Conversely, for every $A \in L(\mathcal{Y})$, the map Φ is recovered from its Choi operator through

$$\Phi(A) = \mathrm{Tr}_{\mathcal{Y}}(J(\Phi)(\mathrm{Id}_{\mathcal{X}} \otimes A^T)). \quad (79)$$

These formulas agree with Watrous [4, Eqs. (2.65)–(2.66)].

We will use the following relation between complete positivity and the Choi operator [4, Thm. 2.22].

Theorem V.1. *Let $\Phi \in T(\mathcal{Y}, \mathcal{X})$. The following statements are equivalent:*

1. $\Phi \in \mathrm{CP}(\mathcal{Y}, \mathcal{X})$;
2. $\Phi \otimes \mathrm{Id}_{L(\mathcal{Y})}$ is positive;

3. $J(\Phi) \in \mathrm{Pos}(\mathcal{X} \otimes \mathcal{Y})$.

The cited theorem is stated for nonzero maps; the zero map also satisfies the three conditions trivially. We will also use the trace-preserving characterization

$$\Phi \text{ is trace-preserving} \iff \mathrm{Tr}_{\mathcal{X}}(J(\Phi)) = \mathrm{Id}_{\mathcal{Y}}, \quad (80)$$

which is given by Watrous [4, Thm. 2.26].

2. Channel induced by a conditional distribution

The conditional distribution $p_{\mathcal{X}|\mathcal{Y}}$ maps an input probability vector $p_{\mathcal{Y}}$ to the output probability vector $p_{\mathcal{X}}$ through the mixture in Eq. (59). The following proposition constructs the corresponding map on operators.

Proposition V.2 (Conditional distributions as quantum channels). *Let $p_{\mathcal{X}|\mathcal{Y}} : \Gamma \rightarrow \mathcal{P}(\Sigma)$ be a conditional distribution, and let $\rho_{\mathcal{X}|\mathcal{Y}}$ be the conditional operator defined in Eq. (71). Then $\rho_{\mathcal{X}|\mathcal{Y}}$ is the Choi operator of a unique quantum channel*

$$\Phi_{\mathcal{X}|\mathcal{Y}} : L(\mathcal{Y}) \rightarrow L(\mathcal{X}). \quad (81)$$

In particular, if $\rho_{\mathcal{Y}}$ represents the probability vector $p_{\mathcal{Y}}$, then

$$\Phi_{\mathcal{X}|\mathcal{Y}}(\rho_{\mathcal{Y}}) = \rho_{\mathcal{X}}, \quad (82)$$

where

$$p_{\mathcal{X}}(x) = \sum_{y \in \Gamma} p_{\mathcal{Y}}(y) p_{\mathcal{X}|\mathcal{Y}}(x|y). \quad (83)$$

Proof. The conditional operator is positive semidefinite and satisfies $\mathrm{Tr}_{\mathcal{X}}(\rho_{\mathcal{X}|\mathcal{Y}}) = \mathrm{Id}_{\mathcal{Y}}$, as shown above. The Choi characterizations of complete positivity and trace preservation therefore imply that it is the Choi operator of a quantum channel. The uniqueness of this channel follows because J is a linear bijection. The inverse Choi–Jamiołkowski formula gives

$$\Phi_{\mathcal{X}|\mathcal{Y}}(A) = \mathrm{Tr}_{\mathcal{Y}}(\rho_{\mathcal{X}|\mathcal{Y}}(\mathrm{Id}_{\mathcal{X}} \otimes A^T)). \quad (84)$$

To understand the action of this channel, it is enough to evaluate it on the matrix units $E_{a,b} \in L(\mathcal{Y})$, for $a, b \in \Gamma$. We have

$$\Phi_{\mathcal{X}|\mathcal{Y}}(E_{a,b}) = \mathrm{Tr}_{\mathcal{Y}}(\rho_{\mathcal{X}|\mathcal{Y}}(\mathrm{Id}_{\mathcal{X}} \otimes E_{a,b}^T)) \quad (85)$$

$$= \mathrm{Tr}_{\mathcal{Y}}(\rho_{\mathcal{X}|\mathcal{Y}}(\mathrm{Id}_{\mathcal{X}} \otimes E_{b,a})) \quad (86)$$

$$= \mathrm{Tr}_{\mathcal{Y}} \left(\sum_{x \in \Sigma} \sum_{y \in \Gamma} p_{\mathcal{X}|\mathcal{Y}}(x|y) \cdot (E_{x,x} \otimes E_{y,y})(\mathrm{Id}_{\mathcal{X}} \otimes E_{b,a}) \right) \quad (87)$$

$$= \mathrm{Tr}_{\mathcal{Y}} \left(\sum_{x \in \Sigma} \sum_{y \in \Gamma} p_{\mathcal{X}|\mathcal{Y}}(x|y) \cdot (E_{x,x} \otimes E_{y,y} E_{b,a}) \right). \quad (88)$$

Here we used $E_{a,b}^T = E_{b,a}$. We also have

$$E_{y,y}E_{b,a} = \delta_{y,b}E_{y,a} = \begin{cases} E_{y,a}, & b = y, \\ 0, & b \neq y. \end{cases} \quad (89)$$

Together with $\text{Tr}(E_{y,a}) = \delta_{y,a}$, this gives

$$\Phi_{X|Y}(E_{a,b}) = \sum_{x \in \Sigma} \sum_{y \in \Gamma} p_{X|Y}(x|y) \text{Tr}_Y(E_{x,x} \otimes \delta_{y,b} E_{y,a}) \quad (90)$$

$$= \sum_{x \in \Sigma} \sum_{y \in \Gamma} p_{X|Y}(x|y) \delta_{y,b} \text{Tr}(E_{y,a}) E_{x,x} \quad (91)$$

$$= \sum_{x \in \Sigma} \sum_{y \in \Gamma} p_{X|Y}(x|y) \delta_{y,b} \delta_{y,a} E_{x,x}. \quad (92)$$

Hence,

$$\Phi_{X|Y}(E_{a,b}) = \begin{cases} \sum_{x \in \Sigma} p_{X|Y}(x|a) E_{x,x}, & a = b, \\ 0, & a \neq b, \end{cases} \quad (93)$$

$$= \begin{cases} \rho_{X|a}, & a = b, \\ 0, & a \neq b. \end{cases} \quad (94)$$

We can now apply this channel to the diagonal state $\rho_Y \in D(\mathcal{Y})$:

$$\Phi_{X|Y}(\rho_Y) = \Phi_{X|Y} \left(\sum_{y \in \Gamma} p_Y(y) E_{y,y} \right) \quad (95)$$

$$= \sum_{y \in \Gamma} p_Y(y) \Phi_{X|Y}(E_{y,y}) \quad (96)$$

$$= \sum_{y \in \Gamma} p_Y(y) \rho_{X|y} \quad (97)$$

$$= \rho_X. \quad (98)$$

The expression in Eq. (97) is exactly the mixture in Eq. (59). Thus, on diagonal states, $\Phi_{X|Y}$ reproduces the stochastic map associated with the conditional distribution $p_{X|Y}$. $\square$

VI. QUANTUM BAYESIAN NETWORKS

This section studies the local factors and channel structure needed for quantum Bayesian networks. We do not consider Bayesian inference here; our aim is to express the classical, quantum, and hybrid factors associated with a directed acyclic graph within a common operator and tensor framework.

A. Classical Bayesian networks in operator form

We first apply the conditional-operator representation of Section V to classical Bayesian networks. We use the

standard Bayesian-network factorization and its associated conditional-independence relations [1, Sec. 11.2.3, Eq. (11.3)]. The two examples below introduce the parent and child configurations used in the subsequent operator and tensor calculations.

1. One child with two parent registers

Let us consider the registers X_1, Y_1, Y_2 with the Bayesian network shown in Fig. 1.

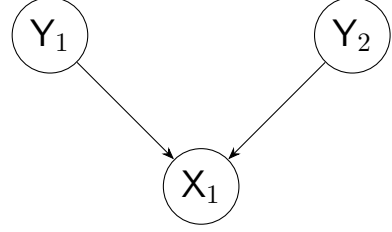


FIG. 1. Bayesian network with two parent registers and one child register.

We denote the associated spaces by $\mathcal{X}_1 = \mathbb{C}^{\Sigma_1}$ and $\mathcal{Y}_i = \mathbb{C}^{\Gamma_i}$ for $i \in \{1, 2\}$. To avoid repeating the two parent registers in every expression, let

$$\mathbf{Y} = (Y_1, Y_2), \quad \mathbf{\Gamma} = \Gamma_1 \times \Gamma_2, \quad \mathcal{Y} = \mathcal{Y}_1 \otimes \mathcal{Y}_2. \quad (99)$$

For $\mathbf{y} = (y_1, y_2) \in \mathbf{\Gamma}$, we also write

$$E_{\mathbf{y}} = E_{y_1, y_1} \otimes E_{y_2, y_2}. \quad (100)$$

The conditional distribution can then be viewed as the pointwise function

$$p_{X_1|\mathbf{Y}} : \Sigma_1 \times \mathbf{\Gamma} \rightarrow [0, 1], \quad (101)$$

where $p_{X_1|\mathbf{Y}}(x|\mathbf{y}) = p_{X_1|Y_1, Y_2}(x|y_1, y_2)$.

The corresponding Choi state is

$$\rho_{X_1|\mathbf{Y}} = \sum_{x \in \Sigma_1} \sum_{\mathbf{y} \in \mathbf{\Gamma}} p_{X_1|\mathbf{Y}}(x|\mathbf{y}) (E_{x,x} \otimes E_{\mathbf{y}}), \quad (102)$$

where $\rho_{X_1|\mathbf{Y}} \equiv \rho_{X_1|Y_1, Y_2}$. Its normalization is

$$\text{Tr}_{\mathcal{X}_1}(\rho_{X_1|\mathbf{Y}}) = \text{Id}_{\mathcal{Y}}. \quad (103)$$

It therefore defines the quantum stochastic map

$$\Phi_{X_1|\mathbf{Y}} : L(\mathcal{Y}) \rightarrow L(\mathcal{X}_1) \quad (104)$$

given by

$$\Phi_{X_1|\mathbf{Y}}(A) = \text{Tr}_{\mathcal{Y}}(\rho_{X_1|\mathbf{Y}}(\text{Id}_{\mathcal{X}_1} \otimes A^T)). \quad (105)$$

Since Y_1 and Y_2 are independent root registers, their joint state factorizes as

$$\rho_{\mathbf{Y}} = \rho_{Y_1} \otimes \rho_{Y_2} = \bigotimes_{i=1}^2 \left(\sum_{y_i \in \Gamma_i} p_{Y_i}(y_i) E_{y_i, y_i} \right). \quad (106)$$

Applying the map to the joint state of the parent registers gives the state of the child register:

$$\Phi_{X_1|\mathbf{Y}}(\rho_{\mathbf{Y}}) = \rho_{X_1}. \quad (107)$$

2. Two child registers with one parent

We now consider the registers $\mathbf{X}_1, \mathbf{X}_2, \mathbf{Y}$ with the Bayesian network shown in Fig. 2.

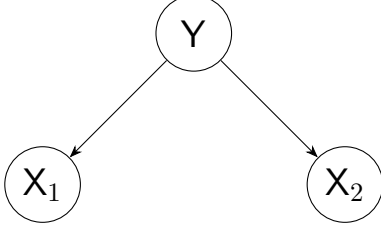


FIG. 2. Bayesian network with one parent register and two child registers.

We denote the associated spaces by $\mathcal{X}_i = \mathbb{C}^{\Sigma_i}$ for $i \in \{1, 2\}$ and $\mathcal{Y} = \mathbb{C}^\Gamma$. As in the preceding example, it is useful to group the two child registers:

$$\mathbf{X} = (\mathbf{X}_1, \mathbf{X}_2), \quad \Sigma = \Sigma_1 \times \Sigma_2, \quad \mathcal{X} = \mathcal{X}_1 \otimes \mathcal{X}_2. \quad (108)$$

For $\mathbf{x} = (x_1, x_2) \in \Sigma$, define

$$E_{\mathbf{x}} = E_{x_1, x_1} \otimes E_{x_2, x_2}. \quad (109)$$

The conditional operator is then

$$\rho_{\mathbf{X}|\mathbf{Y}} = \sum_{\mathbf{x} \in \Sigma} \sum_{y \in \Gamma} p_{\mathbf{X}|\mathbf{Y}}(\mathbf{x}|y) (E_{\mathbf{x}} \otimes E_{y,y}), \quad (110)$$

where $\rho_{\mathbf{X}|\mathbf{Y}} \equiv \rho_{\mathbf{X}_1, \mathbf{X}_2|\mathbf{Y}}$ and $p_{\mathbf{X}|\mathbf{Y}}(\mathbf{x}|y) = p_{\mathbf{X}_1, \mathbf{X}_2|\mathbf{Y}}(x_1, x_2|y)$. We are also interested in the two conditional Choi states $\rho_{\mathbf{X}_1|\mathbf{Y}}$ and $\rho_{\mathbf{X}_2|\mathbf{Y}}$. They are obtained by reducing $\rho_{\mathbf{X}|\mathbf{Y}}$. For example, the partial trace over $\mathcal{X}_2$ is

$$\text{Tr}_{\mathcal{X}_2} = \text{Id}_{L(\mathcal{X}_1)} \otimes \text{Tr} \otimes \text{Id}_{L(\mathcal{Y})}. \quad (111)$$

Applying it gives

$$\text{Tr}_{\mathcal{X}_2}(\rho_{\mathbf{X}|\mathbf{Y}}) = \sum_{\mathbf{x} \in \Sigma} \sum_{y \in \Gamma} p_{\mathbf{X}|\mathbf{Y}}(\mathbf{x}|y) (E_{x_1, x_1} \otimes E_{y,y}). \quad (112)$$

Using the classical conditional marginalization

$$p_{\mathbf{X}_1|\mathbf{Y}}(x_1|y) = \sum_{x_2 \in \Sigma_2} p_{\mathbf{X}|\mathbf{Y}}((x_1, x_2)|y), \quad (113)$$

we obtain

$$\text{Tr}_{\mathcal{X}_2}(\rho_{\mathbf{X}|\mathbf{Y}}) = \sum_{x_1 \in \Sigma_1} \sum_{y \in \Gamma} p_{\mathbf{X}_1|\mathbf{Y}}(x_1|y) (E_{x_1, x_1} \otimes E_{y,y}) \quad (114)$$

$$= \rho_{\mathbf{X}_1|\mathbf{Y}}. \quad (115)$$

Using the same argument, we have

$$\text{Tr}_{\mathcal{X}_1}(\rho_{\mathbf{X}|\mathbf{Y}}) = \rho_{\mathbf{X}_2|\mathbf{Y}}. \quad (116)$$

These reductions are the operator analogue of conditional marginalization.

The three Choi states live in different operator spaces:

$$\rho_{\mathbf{X}_1|\mathbf{Y}} \in L(\mathcal{X}_1 \otimes \mathcal{Y}), \quad (117)$$

$$\rho_{\mathbf{X}_2|\mathbf{Y}} \in L(\mathcal{X}_2 \otimes \mathcal{Y}), \quad (118)$$

$$\rho_{\mathbf{X}|\mathbf{Y}} \in L(\mathcal{X} \otimes \mathcal{Y}). \quad (119)$$

Their corresponding stochastic maps satisfy

$$\begin{aligned} \Phi_{\mathbf{X}_i|\mathbf{Y}} &\in T(\mathcal{Y}, \mathcal{X}_i), \quad i \in \{1, 2\}, \\ \Phi_{\mathbf{X}|\mathbf{Y}} &\in T(\mathcal{Y}, \mathcal{X}), \end{aligned} \quad (120)$$

and

$$\Phi_{\mathbf{X}_i|\mathbf{Y}}(\rho_{\mathbf{Y}}) = \rho_{\mathbf{X}_i}, \quad i \in \{1, 2\}, \quad (121)$$

$$\Phi_{\mathbf{X}|\mathbf{Y}}(\rho_{\mathbf{Y}}) = \rho_{\mathbf{X}}. \quad (122)$$

Since $\Phi_{\mathbf{X}|\mathbf{Y}}$ gives the joint state of the two child registers, we can compose it with a partial trace to obtain either marginal state. For example,

$$\text{Tr}_{\mathcal{X}_1} \circ \Phi_{\mathbf{X}|\mathbf{Y}} : L(\mathcal{Y}) \rightarrow L(\mathcal{X}_2), \quad (123)$$

and therefore

$$(\text{Tr}_{\mathcal{X}_1} \circ \Phi_{\mathbf{X}|\mathbf{Y}})(\rho_{\mathbf{Y}}) = \rho_{\mathbf{X}_2}. \quad (124)$$

We call this composition a *marginal quantum stochastic map* and denote it by

$$\Phi_{\mathbf{X}|\mathbf{Y}}^{\mathcal{X}_1} := \text{Tr}_{\mathcal{X}_1} \circ \Phi_{\mathbf{X}|\mathbf{Y}}. \quad (125)$$

Its full form is

$$\Phi_{\mathbf{X}|\mathbf{Y}}^{\mathcal{X}_1}(A) = \text{Tr}_{\mathcal{X}_1} \left(\text{Tr}_{\mathcal{Y}}(\rho_{\mathbf{X}|\mathbf{Y}}(\text{Id}_{\mathcal{X}} \otimes A^T)) \right). \quad (126)$$

The same construction applies when $\mathcal{X}_2$ is traced out.

A natural question is whether the two conditional operators can be combined to recover $\rho_{\mathbf{X}|\mathbf{Y}}$. In this classical setting the answer is yes. The graph implies the conditional independence $\mathbf{X}_1 \perp \mathbf{X}_2 \mid \mathbf{Y}$, and therefore

$$p_{\mathbf{X}|\mathbf{Y}}(\mathbf{x}|y) = p_{\mathbf{X}_1|\mathbf{Y}}(x_1|y) p_{\mathbf{X}_2|\mathbf{Y}}(x_2|y). \quad (127)$$

We first lift each conditional operator to $L(\mathcal{X} \otimes \mathcal{Y})$ by defining

$$\sigma_{\mathbf{X}_1|\mathbf{Y}} = \sum_{x_1 \in \Sigma_1} \sum_{y \in \Gamma} p_{\mathbf{X}_1|\mathbf{Y}}(x_1|y) (E_{x_1, x_1} \otimes \text{Id}_{\mathcal{X}_2} \otimes E_{y,y}), \quad (128)$$

and

$$\sigma_{\mathbf{X}_2|\mathbf{Y}} = \sum_{x_2 \in \Sigma_2} \sum_{y' \in \Gamma} p_{\mathbf{X}_2|\mathbf{Y}}(x_2|y') (\text{Id}_{\mathcal{X}_1} \otimes E_{x_2, x_2} \otimes E_{y', y'}). \quad (129)$$

These operators are diagonal, so they commute. Their matrix product gives

$$\sigma_{\mathbf{X}_1|\mathbf{Y}}\sigma_{\mathbf{X}_2|\mathbf{Y}} = \sum_{\mathbf{x} \in \Sigma} \sum_{y, y' \in \Gamma} p_{\mathbf{X}_1|\mathbf{Y}}(x_1|y)p_{\mathbf{X}_2|\mathbf{Y}}(x_2|y') \cdot (E_{\mathbf{x}} \otimes E_{y,y}E_{y',y'}) \quad (130)$$

$$= \sum_{\mathbf{x} \in \Sigma} \sum_{y, y' \in \Gamma} p_{\mathbf{X}_1|\mathbf{Y}}(x_1|y)p_{\mathbf{X}_2|\mathbf{Y}}(x_2|y') \cdot (E_{\mathbf{x}} \otimes \delta_{y,y'}E_{y',y'}) \quad (131)$$

$$= \sum_{\mathbf{x} \in \Sigma} \sum_{y \in \Gamma} p_{\mathbf{X}_1|\mathbf{Y}}(x_1|y)p_{\mathbf{X}_2|\mathbf{Y}}(x_2|y) \cdot (E_{\mathbf{x}} \otimes E_{y,y}) \quad (132)$$

$$= \sum_{\mathbf{x} \in \Sigma} \sum_{y \in \Gamma} p_{\mathbf{X}|\mathbf{Y}}(\mathbf{x}|y) (E_{\mathbf{x}} \otimes E_{y,y}) \quad (133)$$

$$= \rho_{\mathbf{X}|\mathbf{Y}}. \quad (134)$$

Hence, after placing the two operators in the common space $L(\mathcal{X} \otimes \mathcal{Y})$, their matrix product recovers the joint conditional Choi state:

$$\sigma_{\mathbf{X}_1|\mathbf{Y}}\sigma_{\mathbf{X}_2|\mathbf{Y}} = \rho_{\mathbf{X}|\mathbf{Y}}. \quad (135)$$

a. Factor products. The preceding calculation is the operator representation of the product of two factors [1, Sec. 11.3.1]. Let $A : \Sigma_1 \times \Gamma \rightarrow \mathbb{R}$ and $B : \Sigma_2 \times \Gamma \rightarrow \mathbb{R}$ be factors with

$$\text{Var}(A) = \{\mathbf{X}_1, \mathbf{Y}\}, \quad \text{Var}(B) = \{\mathbf{X}_2, \mathbf{Y}\}. \quad (136)$$

Set $\Omega = \Sigma_1 \times \Sigma_2 \times \Gamma$. Their product is the factor

$$C : \Omega \rightarrow \mathbb{R}. \quad (137)$$

For every joint assignment $\mathbf{z} = (x_1, x_2, y) \in \Omega$, let $\mathbf{z}'$ and $\mathbf{z}''$ denote its restrictions to $\text{Var}(A)$ and $\text{Var}(B)$, respectively. The product factor is then defined by

$$C(\mathbf{z}) = A(\mathbf{z}')B(\mathbf{z}''). \quad (138)$$

If

$$E_{\mathbf{z}} = E_{\mathbf{x}} \otimes E_{y,y}, \quad (139)$$

the diagonal operator associated with C is

$$\begin{aligned} R_C &= \sum_{\mathbf{z} \in \Omega} A(\mathbf{z}')B(\mathbf{z}'')E_{\mathbf{z}} \\ &= \sum_{\mathbf{z} \in \Omega} C(\mathbf{z})E_{\mathbf{z}}. \end{aligned} \quad (140)$$

For the two-child network in Fig. 2, we take $A = p_{\mathbf{X}_1|\mathbf{Y}}$ and $B = p_{\mathbf{X}_2|\mathbf{Y}}$. Conditional independence gives $C = p_{\mathbf{X}|\mathbf{Y}}$, so that $R_C = \rho_{\mathbf{X}|\mathbf{Y}}$, as shown explicitly in Eq. (132).

3. Conditional Choi operators for compound registers

We can now introduce the notation for an arbitrary collection of child and parent registers. Let $\mathbf{X}_1, \dots, \mathbf{X}_m$

be the child registers and $\mathbf{Y}_1, \dots, \mathbf{Y}_n$ the parent registers. We write

$$\mathbf{X} = (\mathbf{X}_1, \dots, \mathbf{X}_m), \quad \mathbf{Y} = (\mathbf{Y}_1, \dots, \mathbf{Y}_n). \quad (141)$$

For each register, let $\mathcal{X}_i = \mathbb{C}^{\Sigma_i}$ for $i \in \{1, \dots, m\}$ and $\mathcal{Y}_j = \mathbb{C}^{\Gamma_j}$ for $j \in \{1, \dots, n\}$. The compound alphabets and spaces are

$$\Sigma = \Sigma_1 \times \dots \times \Sigma_m, \quad \mathcal{X} = \bigotimes_{i=1}^m \mathcal{X}_i, \quad (142)$$

$$\Gamma = \Gamma_1 \times \dots \times \Gamma_n, \quad \mathcal{Y} = \bigotimes_{j=1}^n \mathcal{Y}_j. \quad (143)$$

For realizations of these registers, we use

$$\mathbf{x} = (x_1, \dots, x_m) \in \Sigma, \quad \mathbf{y} = (y_1, \dots, y_n) \in \Gamma, \quad (144)$$

and define

$$E_{\mathbf{x}} = \bigotimes_{i=1}^m E_{x_i, x_i}, \quad E_{\mathbf{y}} = \bigotimes_{j=1}^n E_{y_j, y_j}. \quad (145)$$

The conditional distribution

$$p_{\mathbf{X}|\mathbf{Y}} : \Sigma \times \Gamma \rightarrow [0, 1] \quad (146)$$

is therefore represented by the conditional Choi state

$$\rho_{\mathbf{X}|\mathbf{Y}} = \sum_{\mathbf{x} \in \Sigma} \sum_{\mathbf{y} \in \Gamma} p_{\mathbf{X}|\mathbf{Y}}(\mathbf{x}|\mathbf{y}) (E_{\mathbf{x}} \otimes E_{\mathbf{y}}). \quad (147)$$

It belongs to $L(\mathcal{X} \otimes \mathcal{Y})$ and satisfies

$$\text{Tr}_{\mathcal{X}}(\rho_{\mathbf{X}|\mathbf{Y}}) = \text{Id}_{\mathcal{Y}}. \quad (148)$$

B. Quantum Channel Networks and Q-Factors

We now consider networks composed only of quantum registers and quantum channels. These networks form the purely quantum fragment of the Q-factor framework introduced by Di Guardia, Ehrhard, and Faggian [3, Defs. 13 and 19]. In the absence of classical variables, a Q-factor is a positive operator. When its partial trace over the output space is the identity on the input space, it is the Choi operator of a quantum channel. The networks studied here are therefore quantum channel networks rather than full quantum Bayesian networks.

1. Quantum registers and channel Q-factors

We use the register conventions introduced in Section IV A 2. For a quantum register $\mathbf{Q}$, its associated complex Euclidean space will be denoted by $\mathcal{H}_{\mathbf{Q}}$. In particular, if $\mathbf{Q} = (q_1, \dots, q_n)$ is an n -qubit register, then

$$\mathcal{H}_{\mathbf{Q}} = \bigotimes_{r=1}^n \mathcal{H}_{q_r} \cong \mathbb{C}^{2^n}. \quad (149)$$

We also use bra-ket notation whenever a basis representation is required. If $\{|a\rangle\}$ is the chosen basis, then $|a\rangle\langle b|$ denotes the corresponding matrix unit; in particular, for $\mathcal{X} = \mathbb{C}^\Sigma$, we identify $|a\rangle = e_a$ and $|a\rangle\langle b| = E_{a,b}$.

2. A single-channel network

Consider the following channel network:

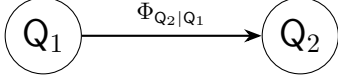


FIG. 3. A single-channel network from Q_1 to Q_2 .

We associate a Q-factor with each register:

1. The initial factor is the state $\rho_{Q_1} \in D(\mathcal{H}_{Q_1})$.
2. The factor associated with Q_2 is

$$\rho_{Q_2|Q_1} \in \text{Pos}(\mathcal{H}_{Q_2} \otimes \mathcal{H}_{Q_1}), \quad (150)$$

with

$$\text{Tr}_{\mathcal{H}_{Q_2}}(\rho_{Q_2|Q_1}) = \text{Id}_{\mathcal{H}_{Q_1}}. \quad (151)$$

The second factor is positive and satisfies the partial-trace normalization above. It is therefore the Choi operator of the unique quantum channel

$$\Phi_{Q_2|Q_1} : L(\mathcal{H}_{Q_1}) \rightarrow L(\mathcal{H}_{Q_2}). \quad (152)$$

When the input is ρ_{Q_1} , the resulting state is

$$\Phi_{Q_2|Q_1}(\rho_{Q_1}) = \rho_{Q_2}. \quad (153)$$

Using the Choi action formula established earlier, the channel is given by [4, Eqs. (2.65)–(2.66)]

$$\Phi_{Q_2|Q_1}(\cdot) = \text{Tr}_{\mathcal{H}_{Q_1}}(\rho_{Q_2|Q_1}(\text{Id}_{\mathcal{H}_{Q_2}} \otimes (\cdot)^T)). \quad (154)$$

Thus, one first transposes the input operator in the chosen basis. The operator

$$\text{Id}_{\mathcal{H}_{Q_2}} \otimes \rho_{Q_1}^T \quad (155)$$

lifts $\rho_{Q_1}^T$ to $L(\mathcal{H}_{Q_2} \otimes \mathcal{H}_{Q_1})$. It can then be multiplied by the Choi operator:

$$\rho_{Q_2|Q_1}(\text{Id}_{\mathcal{H}_{Q_2}} \otimes \rho_{Q_1}^T) \in L(\mathcal{H}_{Q_2} \otimes \mathcal{H}_{Q_1}). \quad (156)$$

Finally, tracing out $\mathcal{H}_{Q_1}$ gives the output density operator ρ_{Q_2} .

3. Block-matrix form of the Choi action

Let $\mathcal{H}_1 = \mathbb{C}^2$ and fix the computational basis $\{|0\rangle, |1\rangle\}$. A normalized pure state $|\psi\rangle \in \mathcal{H}_1$ determines the density operator $\rho = |\psi\rangle\langle\psi|$.

Example VI.1. Write

$$|\psi\rangle = \sum_{r=0}^1 c_r |r\rangle = c_0 |0\rangle + c_1 |1\rangle, \quad (157)$$

where $|c_0|^2 + |c_1|^2 = 1$. The corresponding density operator is

$$\begin{aligned} \rho &= (c_0 |0\rangle + c_1 |1\rangle)(c_0^* \langle 0| + c_1^* \langle 1|) \\ &= \begin{pmatrix} |c_0|^2 & c_0 c_1^* \\ c_1 c_0^* & |c_1|^2 \end{pmatrix}. \end{aligned} \quad (158)$$

For the remainder of the example, we regard ρ as the transpose of an input state $\gamma \in D(\mathcal{H}_1)$, so that $\rho = \gamma^T$. Let $\mathcal{H}_2 \cong \mathbb{C}^2$ be another qubit space and define $\hat{\rho} \in L(\mathcal{H}_2 \otimes \mathcal{H}_1)$ by

$$\begin{aligned} \hat{\rho} &= \text{Id}_{\mathcal{H}_2} \otimes \rho \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \otimes \begin{pmatrix} |c_0|^2 & c_0 c_1^* \\ c_1 c_0^* & |c_1|^2 \end{pmatrix} \\ &= \begin{pmatrix} \rho & \mathbf{0} \\ \mathbf{0} & \rho \end{pmatrix} \\ &= \begin{pmatrix} |c_0|^2 & c_0 c_1^* & 0 & 0 \\ c_1 c_0^* & |c_1|^2 & 0 & 0 \\ 0 & 0 & |c_0|^2 & c_0 c_1^* \\ 0 & 0 & c_1 c_0^* & |c_1|^2 \end{pmatrix}. \end{aligned} \quad (159)$$

This is precisely the lifted operator in Eq. (155).

Now let $\sigma \in L(\mathcal{H}_2 \otimes \mathcal{H}_1)$ be the Choi operator

$$\sigma = \begin{pmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{pmatrix}. \quad (160)$$

The entries are symbolic, but they are constrained by

$$\sigma \in \text{Pos}(\mathcal{H}_2 \otimes \mathcal{H}_1), \quad \text{Tr}_{\mathcal{H}_2}(\sigma) = \text{Id}_{\mathcal{H}_1}. \quad (161)$$

In the channel network above, σ plays the role of $\rho_{Q_2|Q_1}$. Decompose it into blocks:

$$\sigma = \begin{pmatrix} A & B \\ C & D \end{pmatrix}, \quad (162)$$

where

$$\begin{aligned} A &= \begin{pmatrix} a & b \\ e & f \end{pmatrix}, & B &= \begin{pmatrix} c & d \\ g & h \end{pmatrix}, \\ C &= \begin{pmatrix} i & j \\ m & n \end{pmatrix}, & D &= \begin{pmatrix} k & l \\ o & p \end{pmatrix}. \end{aligned} \quad (163)$$

Matrix multiplication gives

$$\sigma \hat{\rho} = \begin{pmatrix} A & B \\ C & D \end{pmatrix} \begin{pmatrix} \rho & 0 \\ 0 & \rho \end{pmatrix} = \begin{pmatrix} A\rho & B\rho \\ C\rho & D\rho \end{pmatrix}. \quad (164)$$

The partial trace over $\mathcal{H}_1$ is therefore

$$\text{Tr}_{\mathcal{H}_1}(\sigma \hat{\rho}) = \begin{pmatrix} \text{Tr}(A\rho) & \text{Tr}(B\rho) \\ \text{Tr}(C\rho) & \text{Tr}(D\rho) \end{pmatrix}. \quad (165)$$

For example,

$$\begin{aligned} A\rho &= \begin{pmatrix} a & b \\ e & f \end{pmatrix} \begin{pmatrix} |c_0|^2 & c_0 c_1^* \\ c_1 c_0^* & |c_1|^2 \end{pmatrix} \\ &= \begin{pmatrix} a|c_0|^2 + bc_1 c_0^* & \star \\ \star & ec_0 c_1^* + f|c_1|^2 \end{pmatrix}, \end{aligned} \quad (166)$$

where the entries denoted by $\star$ do not contribute to the trace.

Consequently,

$$\text{Tr}(A\rho) = a|c_0|^2 + bc_1 c_0^* + ec_0 c_1^* + f|c_1|^2. \quad (167)$$

$$\text{Tr}_{\mathcal{H}_1}(\sigma\hat{\rho}) = \begin{pmatrix} a|c_0|^2 + bc_1 c_0^* & c|c_0|^2 + d c_1 c_0^* \\ + e c_0 c_1^* + f|c_1|^2 & + g c_0 c_1^* + h|c_1|^2 \\ i|c_0|^2 + j c_1 c_0^* & k|c_0|^2 + l c_1 c_0^* \\ + m c_0 c_1^* + n|c_1|^2 & + o c_0 c_1^* + p|c_1|^2 \end{pmatrix}. \quad (168)$$

Here, the matrix entry i is an arbitrary complex number and should not be confused with the imaginary unit.

4. Choi action as a tensor contraction

A natural question is whether we really need to perform all the preceding operations in order to obtain the output of the map $\Phi_{Q_2|Q_1}$. We first transpose the input matrix, lift it to the space of the Choi operator so that the matrix multiplication can be performed, and finally apply the partial trace. Is there a more direct way to carry out this computation?

The operator–tensor correspondence introduced in the preliminaries allows us to express the same calculation as a tensor contraction. Accordingly, the operators σ and $\hat{\rho}$ are represented by tensors σ' and $\hat{\rho}'$, respectively. Relative to the factorization into the two subsystems, these are tensors of type $(2, 2)$ (order four):

$$\sigma', \hat{\rho}' \in (\mathcal{H}_2 \otimes \mathcal{H}_1) \otimes (\mathcal{H}_2 \otimes \mathcal{H}_1)^*. \quad (169)$$

Throughout this subsection, we use the standard tensor-calculus convention introduced in the preliminaries: a repeated index is contracted only when it occurs once in an upper position and once in a lower position. To distinguish tensors from the operators they represent, square brackets will be reserved for the component matrix of a tensor. To obtain the matrix representations of σ' and $\hat{\rho}'$, we use the canonical basis

$$\mathcal{B} = \{|i_2 i_1\rangle \langle j_2 j_1| : i_2, i_1, j_2, j_1 \in \{0, 1\}\}. \quad (170)$$

The pairs (i_2, j_2) and (i_1, j_1) correspond to $\mathcal{H}_2$ and $\mathcal{H}_1$, respectively. We set

$$[\sigma']_{\mathcal{B}} = \sigma, \quad [\hat{\rho}']_{\mathcal{B}} = \hat{\rho},$$

where the operators on the right are written in this canonical basis. Thus,

$$\sigma' = \sum_{i_2, i_1, j_2, j_1} (\sigma')^{i_2 i_1}_{j_2 j_1} |i_2 i_1\rangle \langle j_2 j_1|, \quad (171)$$

and its matrix representation is

$$[\sigma']_{\mathcal{B}} = \begin{pmatrix} (\sigma')^{00}_{00} & (\sigma')^{00}_{01} & (\sigma')^{00}_{10} & (\sigma')^{00}_{11} \\ (\sigma')^{01}_{00} & (\sigma')^{01}_{01} & (\sigma')^{01}_{10} & (\sigma')^{01}_{11} \\ (\sigma')^{10}_{00} & (\sigma')^{10}_{01} & (\sigma')^{10}_{10} & (\sigma')^{10}_{11} \\ (\sigma')^{11}_{00} & (\sigma')^{11}_{01} & (\sigma')^{11}_{10} & (\sigma')^{11}_{11} \end{pmatrix}. \quad (172)$$

Ordinary matrix multiplication is represented by the upper–lower contraction

$$\varphi^{i_2 i_1}_{m_2 m_1} = (\sigma')^{i_2 i_1}_{j_2 j_1} (\hat{\rho}')^{j_2 j_1}_{m_2 m_1}, \quad (173)$$

or, with the sums displayed,

$$\varphi^{i_2 i_1}_{m_2 m_1} = \sum_{j_2, j_1} (\sigma')^{i_2 i_1}_{j_2 j_1} (\hat{\rho}')^{j_2 j_1}_{m_2 m_1}. \quad (174)$$

The tensor φ therefore satisfies $[\varphi]_{\mathcal{B}} = \sigma\hat{\rho}$. Its full component matrix is

$$[\varphi]_{\mathcal{B}} = \begin{pmatrix} \varphi^{00}_{00} & \varphi^{00}_{01} & \varphi^{00}_{10} & \varphi^{00}_{11} \\ \varphi^{01}_{00} & \varphi^{01}_{01} & \varphi^{01}_{10} & \varphi^{01}_{11} \\ \varphi^{10}_{00} & \varphi^{10}_{01} & \varphi^{10}_{10} & \varphi^{10}_{11} \\ \varphi^{11}_{00} & \varphi^{11}_{01} & \varphi^{11}_{10} & \varphi^{11}_{11} \end{pmatrix} \quad (175)$$

The brown indices refer to $\mathcal{H}_2$, while the green indices refer to $\mathcal{H}_1$. It is useful to view this order-four tensor as a 2×2 array of 2×2 blocks. For fixed $i_2, j_2 \in \{0, 1\}$, define

$$\varphi^{i_2}_{j_2} := \begin{pmatrix} \varphi^{i_2 0}_{j_2 0} & \varphi^{i_2 0}_{j_2 1} \\ \varphi^{i_2 1}_{j_2 0} & \varphi^{i_2 1}_{j_2 1} \end{pmatrix}. \quad (176)$$

$$\begin{pmatrix} \varphi^0_0 & \varphi^0_1 \\ \begin{pmatrix} \varphi^{00}_{00} & \varphi^{00}_{01} \\ \varphi^{01}_{00} & \varphi^{01}_{01} \end{pmatrix} & \begin{pmatrix} \varphi^{00}_{10} & \varphi^{00}_{11} \\ \varphi^{01}_{10} & \varphi^{01}_{11} \end{pmatrix} \\ \varphi^1_0 & \varphi^1_1 \\ \begin{pmatrix} \varphi^{10}_{00} & \varphi^{10}_{01} \\ \varphi^{11}_{00} & \varphi^{11}_{01} \end{pmatrix} & \begin{pmatrix} \varphi^{10}_{10} & \varphi^{10}_{11} \\ \varphi^{11}_{10} & \varphi^{11}_{11} \end{pmatrix} \end{pmatrix}$$

FIG. 4. The tensor φ represented as a 2×2 array of 2×2 matrices.

The reason the representation in Fig. 4 is useful is that the partial trace over $\mathcal{H}_1$ is obtained by taking the ordinary trace of each block $\varphi^{i_2}_{j_2}$. Indeed, for the operator represented by $[\varphi]_{\mathcal{B}}$,

$$(\text{Tr}_{\mathcal{H}_1}(\varphi))^{i_2}_{j_2} = \text{Tr}(\varphi^{i_2}_{j_2}) = \varphi^{i_2 i_1}_{j_2 i_1}. \quad (177)$$

The result is a tensor of type $(1, 1)$ on $\mathcal{H}_2$ and agrees with Eq. (168). Indeed, $[\sigma']_{\mathcal{B}}$ and $[\hat{\rho}']_{\mathcal{B}}$ are the operator matrices in Eqs. (160) and (159). Relative to the canonical basis

$$\mathcal{A} = \{|i_2\rangle \langle j_2| : i_2, j_2 \in \{0, 1\}\} \quad (178)$$

of $L(\mathcal{H}_2)$, one has

$$[\text{Tr}_{\mathcal{H}_1}(\varphi)]_{\mathcal{A}} = \text{Tr}_{\mathcal{H}_1}([\sigma']_{\mathcal{B}}[\hat{\rho}']_{\mathcal{B}}). \quad (179)$$

We can now bypass the lifted matrix. Let

$$\mathcal{A}' = \{|i_1\rangle\langle j_1| : i_1, j_1 \in \{0, 1\}\} \quad (180)$$

be the canonical basis of $L(\mathcal{H}_1)$, and write the components of ρ as $\rho^{i_1}_{j_1}$. Direct contraction with the input indices of σ' defines

$$\chi^{i_2}_{j_2} = (\sigma')^{i_2 i_1}_{j_2 j_1} \rho^{j_1}_{i_1}. \quad (181)$$

The exchanged indices of ρ account for the transpose in the Choi action. More explicitly,

$$(\sigma')^{i_2 i_1}_{j_2 j_1} \rho^{j_1}_{i_1} = \sum_{i_1} \sum_{j_1} (\sigma')^{i_2 i_1}_{j_2 j_1} \rho^{j_1}_{i_1} \quad (182)$$

$$= \sum_{i_1} ((\sigma')^{i_2 i_1}_{j_2 0} \rho^0_{i_1} + (\sigma')^{i_2 i_1}_{j_2 1} \rho^1_{i_1}) \quad (183)$$

$$= (\sigma')^{i_2 0}_{j_2 0} \rho^0_0 + (\sigma')^{i_2 0}_{j_2 1} \rho^1_0 + (\sigma')^{i_2 1}_{j_2 0} \rho^0_1 + (\sigma')^{i_2 1}_{j_2 1} \rho^1_1. \quad (184)$$

For example, the first two entries of the output are

$$\begin{aligned} \chi^0_0 &= (\sigma')^{00}_{00} \rho^0_0 + (\sigma')^{00}_{01} \rho^1_0 \\ &\quad + (\sigma')^{01}_{00} \rho^0_1 + (\sigma')^{01}_{01} \rho^1_1 \end{aligned} \quad (185)$$

$$= a|c_0|^2 + bc_1c_0^* + ec_0c_1^* + f|c_1|^2, \quad (186)$$

$$\begin{aligned} \chi^0_1 &= (\sigma')^{00}_{10} \rho^0_0 + (\sigma')^{00}_{11} \rho^1_0 \\ &\quad + (\sigma')^{01}_{10} \rho^0_1 + (\sigma')^{01}_{11} \rho^1_1 \end{aligned} \quad (187)$$

$$= c|c_0|^2 + dc_1c_0^* + gc_0c_1^* + h|c_1|^2. \quad (188)$$

These are precisely the corresponding entries of Eq. (168). Therefore,

$$\begin{aligned} \chi^{i_2}_{j_2} &= (\text{Tr}_{\mathcal{H}_1}(\sigma\hat{\rho}))^{i_2}_{j_2} \\ &= (\text{Tr}_{\mathcal{H}_1}(\sigma(\text{Id}_{\mathcal{H}_2} \otimes \rho)))^{i_2}_{j_2}. \end{aligned} \quad (189)$$

Proposition VI.1 (Choi action as a tensor contraction). *Let*

$$\Phi_{\mathcal{Q}_2|\mathcal{Q}_1} : L(\mathcal{H}_{\mathcal{Q}_1}) \rightarrow L(\mathcal{H}_{\mathcal{Q}_2}) \quad (190)$$

be a quantum channel with Choi operator $\rho_{\mathcal{Q}_2|\mathcal{Q}_1}$. Let

$$\rho'_{\mathcal{Q}_2|\mathcal{Q}_1} \in (\mathcal{H}_{\mathcal{Q}_2} \otimes \mathcal{H}_{\mathcal{Q}_1}) \otimes (\mathcal{H}_{\mathcal{Q}_2} \otimes \mathcal{H}_{\mathcal{Q}_1})^* \quad (191)$$

be the tensor representing this Choi operator. For $\rho_{\mathcal{Q}_1} \in D(\mathcal{H}_{\mathcal{Q}_1})$, let $\rho'_{\mathcal{Q}_1} \in \mathcal{H}_{\mathcal{Q}_1} \otimes \mathcal{H}_{\mathcal{Q}_1}^*$ represent the transposed input state. If $\mathcal{C}$, $\mathcal{B}$, and $\mathcal{B}'$ are the canonical bases of the corresponding operator spaces, then

$$[\rho'_{\mathcal{Q}_2|\mathcal{Q}_1}]_{\mathcal{C}} = \rho_{\mathcal{Q}_2|\mathcal{Q}_1}, \quad [\rho'_{\mathcal{Q}_1}]_{\mathcal{B}} = \rho_{\mathcal{Q}_1}^T. \quad (192)$$

The tensor $\rho'_{\mathcal{Q}_2} \in \mathcal{H}_{\mathcal{Q}_2} \otimes \mathcal{H}_{\mathcal{Q}_2}^*$ defined by

$$(\rho'_{\mathcal{Q}_2})^{i_2}_{j_2} = (\rho'_{\mathcal{Q}_2|\mathcal{Q}_1})^{i_2 i_1}_{j_2 j_1} (\rho'_{\mathcal{Q}_1})^{j_1}_{i_1} \quad (193)$$

represents the output state. In other words,

$$[\rho'_{\mathcal{Q}_2}]_{\mathcal{B}'} = \Phi_{\mathcal{Q}_2|\mathcal{Q}_1}(\rho_{\mathcal{Q}_1}) = \rho_{\mathcal{Q}_2}. \quad (194)$$

Proof sketch. The inverse Choi–Jamiołkowski formula gives

$$(\Phi_{\mathcal{Q}_2|\mathcal{Q}_1}(\rho_{\mathcal{Q}_1}))^{i_2}_{j_2} = (\rho'_{\mathcal{Q}_2|\mathcal{Q}_1})^{i_2 i_1}_{j_2 j_1} (\rho'_{\mathcal{Q}_1})^{j_1}_{i_1}. \quad (195)$$

This proves the stated contraction identity. For the qubit channel considered in Example VI.1, the same identity can be verified explicitly as follows. Let

$$\hat{\rho}_{\mathcal{Q}_1} \in (\mathcal{H}_{\mathcal{Q}_2} \otimes \mathcal{H}_{\mathcal{Q}_1}) \otimes (\mathcal{H}_{\mathcal{Q}_2} \otimes \mathcal{H}_{\mathcal{Q}_1})^* \quad (196)$$

be the lifted tensor satisfying

$$[\hat{\rho}_{\mathcal{Q}_1}]_{\mathcal{C}} = \text{Id}_{\mathcal{H}_{\mathcal{Q}_2}} \otimes \rho_{\mathcal{Q}_1}^T. \quad (197)$$

Its complete component matrix is

$$[\hat{\rho}_{\mathcal{Q}_1}]_{\mathcal{C}} = \begin{pmatrix} (\hat{\rho}_{\mathcal{Q}_1})^{00}_{00} & (\hat{\rho}_{\mathcal{Q}_1})^{00}_{01} & (\hat{\rho}_{\mathcal{Q}_1})^{00}_{10} & (\hat{\rho}_{\mathcal{Q}_1})^{00}_{11} \\ (\hat{\rho}_{\mathcal{Q}_1})^{01}_{00} & (\hat{\rho}_{\mathcal{Q}_1})^{01}_{01} & (\hat{\rho}_{\mathcal{Q}_1})^{01}_{10} & (\hat{\rho}_{\mathcal{Q}_1})^{01}_{11} \\ (\hat{\rho}_{\mathcal{Q}_1})^{10}_{00} & (\hat{\rho}_{\mathcal{Q}_1})^{10}_{01} & (\hat{\rho}_{\mathcal{Q}_1})^{10}_{10} & (\hat{\rho}_{\mathcal{Q}_1})^{10}_{11} \\ (\hat{\rho}_{\mathcal{Q}_1})^{11}_{00} & (\hat{\rho}_{\mathcal{Q}_1})^{11}_{01} & (\hat{\rho}_{\mathcal{Q}_1})^{11}_{10} & (\hat{\rho}_{\mathcal{Q}_1})^{11}_{11} \end{pmatrix}. \quad (198)$$

Its two off-diagonal blocks satisfy

$$\begin{aligned} \begin{pmatrix} (\hat{\rho}_{\mathcal{Q}_1})^{00}_{10} & (\hat{\rho}_{\mathcal{Q}_1})^{00}_{11} \\ (\hat{\rho}_{\mathcal{Q}_1})^{01}_{10} & (\hat{\rho}_{\mathcal{Q}_1})^{01}_{11} \end{pmatrix} &= \begin{pmatrix} (\hat{\rho}_{\mathcal{Q}_1})^{10}_{00} & (\hat{\rho}_{\mathcal{Q}_1})^{10}_{01} \\ (\hat{\rho}_{\mathcal{Q}_1})^{11}_{00} & (\hat{\rho}_{\mathcal{Q}_1})^{11}_{01} \end{pmatrix} \\ &= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}. \end{aligned} \quad (199)$$

Thus $[\hat{\rho}_{\mathcal{Q}_1}]_{\mathcal{C}}$ is block diagonal. Componentwise,

$$(\hat{\rho}_{\mathcal{Q}_1})^{i_2 i_1}_{j_2 j_1} = \begin{cases} 0, & i_2 \neq j_2, \\ (\rho'_{\mathcal{Q}_1})^{i_1}_{j_1}, & i_2 = j_2. \end{cases} \quad (200)$$

Equivalently,

$$(\hat{\rho}_{\mathcal{Q}_1})^{i_2 i_1}_{j_2 j_1} = \delta^{i_2}_{j_2} (\rho'_{\mathcal{Q}_1})^{i_1}_{j_1}. \quad (201)$$

This is exactly the tensor form of $\text{Id}_{\mathcal{H}_{\mathcal{Q}_2}} \otimes \rho_{\mathcal{Q}_1}^T$. The contraction

$$(\rho'_{\mathcal{Q}_2|\mathcal{Q}_1})^{i_2 i_1}_{j_2 j_1} (\hat{\rho}_{\mathcal{Q}_1})^{j_2 j_1}_{m_2 m_1} \quad (202)$$

is the component form of the matrix product

$$(\rho_{\mathcal{Q}_2|\mathcal{Q}_1} (\text{Id}_{\mathcal{H}_{\mathcal{Q}_2}} \otimes \rho_{\mathcal{Q}_1}^T))_{ij} = (\rho_{\mathcal{Q}_2|\mathcal{Q}_1})_{ik} (\text{Id}_{\mathcal{H}_{\mathcal{Q}_2}} \otimes \rho_{\mathcal{Q}_1}^T)_{kj}. \quad (203)$$

Applying the partial trace over $\mathcal{Q}_1$ identifies m_1 with i_1 :

$$(\rho'_{\mathcal{Q}_2|\mathcal{Q}_1})^{i_2 i_1}_{j_2 j_1} (\hat{\rho}_{\mathcal{Q}_1})^{j_2 j_1}_{m_2 i_1}. \quad (204)$$

Hence

$$(\rho'_{\mathcal{Q}_2})^{i_2}_{m_2} = (\rho'_{\mathcal{Q}_2|\mathcal{Q}_1})^{i_2 i_1}_{j_2 j_1} (\hat{\rho}_{\mathcal{Q}_1})^{j_2 j_1}_{m_2 i_1}. \quad (205)$$

Substituting expression (201) gives

$$\begin{aligned} &(\rho'_{\mathcal{Q}_2|\mathcal{Q}_1})^{i_2 i_1}_{j_2 j_1} (\hat{\rho}_{\mathcal{Q}_1})^{j_2 j_1}_{m_2 m_1} \\ &= (\rho'_{\mathcal{Q}_2|\mathcal{Q}_1})^{i_2 i_1}_{j_2 j_1} \delta^{j_2}_{m_2} (\rho'_{\mathcal{Q}_1})^{j_1}_{m_1} \\ &= (\rho'_{\mathcal{Q}_2|\mathcal{Q}_1})^{i_2 i_1}_{m_2 j_1} (\rho'_{\mathcal{Q}_1})^{j_1}_{m_1}. \end{aligned} \quad (206)$$

Define

$$\varphi^{i_2 i_1}_{m_2 m_1} = (\rho'_{Q_2|Q_1})^{i_2 i_1}_{m_2 j_1} (\rho'_{Q_1})^{j_1}_{m_1}. \quad (207)$$

Taking the partial trace gives

$$(\text{Tr}_{\mathcal{H}_{Q_1}}(\varphi))^{i_2}_{m_2} = (\rho'_{Q_2|Q_1})^{i_2 i_1}_{m_2 j_1} (\rho'_{Q_1})^{j_1}_{i_1}. \quad (208)$$

Therefore,

$$\begin{aligned} (\rho'_{Q_2})^{i_2}_{m_2} &= (\rho'_{Q_2|Q_1})^{i_2 i_1}_{j_2 j_1} (\hat{\rho}_{Q_1})^{j_2 j_1}_{m_2 i_1} \\ &= (\rho'_{Q_2|Q_1})^{i_2 i_1}_{m_2 j_1} (\rho'_{Q_1})^{j_1}_{i_1}. \end{aligned} \quad (209)$$

□

The Choi action is therefore obtained directly by contracting the indices of the input register. This single contraction encodes both the multiplication by the lifted transposed input and the subsequent partial trace.

5. Composition of channel Q-factors

Channel composition can also be expressed directly in terms of Choi operators through the link product [12]. Here we work with tensor components in order to keep the contraction over the intermediate register explicit.

Consider the chain of two quantum channels shown in Fig. 5.

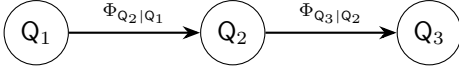


FIG. 5. A two-channel chain.

The associated Q-factors are

1. $\rho_{Q_1} \in D(\mathcal{H}_{Q_1})$;
2. $\rho_{Q_2|Q_1} \in \text{Pos}(\mathcal{H}_{Q_2} \otimes \mathcal{H}_{Q_1})$, with

$$\text{Tr}_{\mathcal{H}_{Q_2}}(\rho_{Q_2|Q_1}) = \text{Id}_{\mathcal{H}_{Q_1}}; \quad (210)$$

3. $\rho_{Q_3|Q_2} \in \text{Pos}(\mathcal{H}_{Q_3} \otimes \mathcal{H}_{Q_2})$, with

$$\text{Tr}_{\mathcal{H}_{Q_3}}(\rho_{Q_3|Q_2}) = \text{Id}_{\mathcal{H}_{Q_2}}. \quad (211)$$

The last two factors are the Choi operators of the corresponding channels. Their component contraction is the pure-quantum instance of the product of Q-factors [3, Def. 17].

For this example, Q_1 , Q_2 , and Q_3 are qubit registers. Each index i_r, j_r is associated with $\mathcal{H}_{Q_r}$ and takes values in $\{0, 1\}$. The corresponding tensors are

1. $\varrho_{Q_1} \in \mathcal{H}_{Q_1} \otimes \mathcal{H}_{Q_1}^*$, with components $(\varrho_{Q_1})^{i_1}_{j_1}$;
2. $\varrho_{Q_2|Q_1} \in (\mathcal{H}_{Q_2} \otimes \mathcal{H}_{Q_1}) \otimes (\mathcal{H}_{Q_2} \otimes \mathcal{H}_{Q_1})^*$, with components $(\varrho_{Q_2|Q_1})^{i_2 i_1}_{j_2 j_1}$;

3. $\varrho_{Q_3|Q_2} \in (\mathcal{H}_{Q_3} \otimes \mathcal{H}_{Q_2}) \otimes (\mathcal{H}_{Q_3} \otimes \mathcal{H}_{Q_2})^*$, with components $(\varrho_{Q_3|Q_2})^{i_3 i_2}_{j_3 j_2}$.

Let $\mathcal{B}^{(r)}$ denote the canonical basis of $L(\mathcal{H}_{Q_r})$, let $\mathcal{A}$ denote the canonical basis of $L(\mathcal{H}_{Q_2} \otimes \mathcal{H}_{Q_1})$, and let $\mathcal{A}'$ denote the canonical basis of $L(\mathcal{H}_{Q_3} \otimes \mathcal{H}_{Q_2})$. We set

$$\begin{aligned} [\varrho_{Q_1}]_{\mathcal{B}^{(1)}} &= \rho_{Q_1}^T, \\ [\varrho_{Q_2|Q_1}]_{\mathcal{A}} &= \rho_{Q_2|Q_1}, \quad [\varrho_{Q_3|Q_2}]_{\mathcal{A}'} = \rho_{Q_3|Q_2}. \end{aligned} \quad (212)$$

and

$$[\varrho_{Q_2}]_{\mathcal{B}^{(2)}} = \rho_{Q_2}, \quad [\varrho_{Q_3}]_{\mathcal{B}^{(3)}} = \rho_{Q_3}. \quad (213)$$

Before writing the component contraction, consider the two channels

$$\Phi_{Q_2|Q_1}, \quad \Phi_{Q_3|Q_2}. \quad (214)$$

Their composition satisfies

$$(\Phi_{Q_3|Q_2} \circ \Phi_{Q_2|Q_1})(\rho_{Q_1}) = \rho_{Q_3}. \quad (215)$$

Applying the Choi action formula successively gives

$$\begin{aligned} \rho_{Q_2} &= \text{Tr}_{\mathcal{H}_{Q_1}}(\rho_{Q_2|Q_1} (\text{Id}_{\mathcal{H}_{Q_2}} \otimes \rho_{Q_1}^T)), \\ (\Phi_{Q_3|Q_2} \circ \Phi_{Q_2|Q_1})(\rho_{Q_1}) &= \text{Tr}_{\mathcal{H}_{Q_2}}(\rho_{Q_3|Q_2} (\text{Id}_{\mathcal{H}_{Q_3}} \otimes \rho_{Q_2}^T)). \end{aligned} \quad (216)$$

Can we express this composition with tensors? As we saw in Section VIB 4,

$$(\varrho_{Q_2})^{i_2}_{j_2} = (\varrho_{Q_1})^{j_1}_{i_1} (\varrho_{Q_2|Q_1})^{i_2 i_1}_{j_2 j_1}. \quad (217)$$

But can we find ϱ_{Q_3} while keeping our definition of ϱ_{Q_2} ? The composition of the maps in Eq. (216) would formally take the form

$$(\varrho_{Q_3})^{i_3}_{j_3} = (\varrho_{Q_2|Q_1})^{i_2 i_1}_{j_2 j_1} (\varrho_{Q_3|Q_2})^{i_3 i_2}_{j_3 j_2} (\varrho_{Q_1})^{j_1}_{i_1}. \quad (218)$$

Using Eq. (217), this would reduce to

$$(\varrho_{Q_3})^{i_3}_{j_3} = (\varrho_{Q_2})^{i_2}_{j_2} (\varrho_{Q_3|Q_2})^{i_3 i_2}_{j_3 j_2}. \quad (219)$$

At this point, however, we encounter a problem. In Eq. (219), i_2 occurs twice in the upper position and j_2 twice in the lower position. Therefore, this is not a tensor contraction according to the standard upper-lower convention used so far. Moreover, if the repeated indices are summed directly, the expression is not covariant under arbitrary changes of basis. This difficulty comes from the transpose of the intermediate state in the Choi action.

To retain our definition of ϱ_{Q_2} and compose the channel factors directly, we now fix an orthonormal basis for each register, taking the computational basis for qubit registers [7, Secs. 6.1–6.3]. We retain the mixed placement of upper and lower indices to indicate row and column indices, but from this point onward a repeated register index is summed over even when both occurrences are upper indices or both are lower indices. With this convention, the two preceding expressions are valid component

identities in these bases. In particular, the contraction in Eq. (219) absorbs the transpose of the intermediate state.

To see explicitly how the transpose is encoded, define $\varsigma_{Q_1} \in \mathcal{H}_{Q_1} \otimes \mathcal{H}_{Q_1}^*$ by

$$[\varsigma_{Q_1}]_{\mathcal{B}^{(1)}} = \rho_{Q_1}. \quad (220)$$

Then

$$(\varrho_{Q_1})^{j_1}_{i_1} = (\varsigma_{Q_1})^{i_1}_{j_1}, \quad (221)$$

and Eq. (217) becomes

$$(\varrho_{Q_2})^{i_2}_{j_2} = (\varsigma_{Q_1})^{i_1}_{j_1} (\varrho_{Q_2|Q_1})^{i_2 i_1}_{j_2 j_1}. \quad (222)$$

This has exactly the same contraction pattern as Eq. (219).

Similarly, define $\xi_{Q_2} \in \mathcal{H}_{Q_2} \otimes \mathcal{H}_{Q_2}^*$ by

$$[\xi_{Q_2}]_{\mathcal{B}^{(2)}} = \rho_{Q_2}^T. \quad (223)$$

Then

$$(\varrho_{Q_2})^{i_2}_{j_2} = (\xi_{Q_2})^{j_2}_{i_2}, \quad (224)$$

so that

$$(\varrho_{Q_3})^{i_3}_{j_3} = (\varrho_{Q_2})^{i_2}_{j_2} (\varrho_{Q_3|Q_2})^{i_3 i_2}_{j_3 j_2} \quad (225)$$

$$= (\xi_{Q_2})^{j_2}_{i_2} (\varrho_{Q_3|Q_2})^{i_3 i_2}_{j_3 j_2}. \quad (226)$$

The last line is the corresponding upper-lower tensor contraction:

$$(\varrho_{Q_3})^{i_3}_{j_3} = (\xi_{Q_2})^{j_2}_{i_2} (\varrho_{Q_3|Q_2})^{i_3 i_2}_{j_3 j_2}. \quad (227)$$

One could instead define the initial tensor directly by

$$[\varrho_{Q_1}]_{\mathcal{B}^{(1)}} = \rho_{Q_1}. \quad (228)$$

With this convention, the correct component formula would then be

$$(\varrho_{Q_1})^{i_1}_{j_1} (\varrho_{Q_2|Q_1})^{i_2 i_1}_{j_2 j_1} = (\varrho_{Q_2})^{i_2}_{j_2}. \quad (229)$$

By contrast, with this new definition the expression

$$(\varrho_{Q_1})^{j_1}_{i_1} (\varrho_{Q_2|Q_1})^{i_2 i_1}_{j_2 j_1} = (\varrho_{Q_2})^{i_2}_{j_2} \quad (230)$$

would insert an additional transpose and is therefore incorrect.

We henceforth adopt this latter convention, in which a state tensor represents the density operator itself rather than its transpose. We now revisit the composition of quantum channels in tensor form. The following proposition may be viewed as a generalization of Proposition VI.1.

Proposition VI.2 (Quantum-channel composition as tensor contraction). *Let*

$$\Phi_{Q_2|Q_1} : L(\mathcal{H}_{Q_1}) \rightarrow L(\mathcal{H}_{Q_2}) \quad (231)$$

and

$$\Phi_{Q_3|Q_2} : L(\mathcal{H}_{Q_2}) \rightarrow L(\mathcal{H}_{Q_3}) \quad (232)$$

be quantum channels. Let their Choi operators be represented by $\varrho_{Q_2|Q_1}$ and $\varrho_{Q_3|Q_2}$ in the chosen bases.

The Choi operator of the composite channel

$$\Phi_{Q_3|Q_1} = \Phi_{Q_3|Q_2} \circ \Phi_{Q_2|Q_1} \quad (233)$$

is represented by the tensor whose components are

$$(\varrho_{Q_3|Q_1})^{i_3 i_1}_{j_3 j_1} = (\varrho_{Q_2|Q_1})^{i_2 i_1}_{j_2 j_1} (\varrho_{Q_3|Q_2})^{i_3 i_2}_{j_3 j_2}. \quad (234)$$

Here, the indices i_2 and j_2 associated with the intermediate register are summed according to the component convention introduced above.

Consequently, if ϱ_{Q_1} represents ρ_{Q_1} , then

$$(\varrho_{Q_3})^{i_3}_{j_3} = (\varrho_{Q_2|Q_1})^{i_2 i_1}_{j_2 j_1} (\varrho_{Q_3|Q_2})^{i_3 i_2}_{j_3 j_2} (\varrho_{Q_1})^{i_1}_{j_1}. \quad (235)$$

Proof. By definition, the Choi operator of the composite channel is

$$\begin{aligned} \rho_{Q_3|Q_1} &= J(\Phi_{Q_3|Q_2} \circ \Phi_{Q_2|Q_1}) \\ &= \sum_{a,b} (\Phi_{Q_3|Q_2} \circ \Phi_{Q_2|Q_1})(E_{a,b}) \otimes E_{a,b}. \end{aligned} \quad (236)$$

Its component with row indices (i_3, i_1) and column indices (j_3, j_1) is

$$(\varrho_{Q_3|Q_1})^{i_3 i_1}_{j_3 j_1} = \langle i_3 i_1 | \rho_{Q_3|Q_1} | j_3 j_1 \rangle. \quad (237)$$

Substituting the preceding expression for the Choi operator gives

$$\begin{aligned} &(\varrho_{Q_3|Q_1})^{i_3 i_1}_{j_3 j_1} \\ &= \sum_{a,b} \langle i_3 | (\Phi_{Q_3|Q_2} \circ \Phi_{Q_2|Q_1})(E_{a,b}) | j_3 \rangle \\ &\quad \cdot \langle i_1 | E_{a,b} | j_1 \rangle. \end{aligned} \quad (238)$$

Since $E_{a,b} = |a\rangle\langle b|$,

$$\langle i_1 | E_{a,b} | j_1 \rangle = \delta_{i_1,a} \delta_{j_1,b}. \quad (239)$$

The sums over a and b therefore reduce to $a = i_1$ and $b = j_1$, and hence

$$\begin{aligned} &(\varrho_{Q_3|Q_1})^{i_3 i_1}_{j_3 j_1} \\ &= \langle i_3 | \Phi_{Q_3|Q_2} (\Phi_{Q_2|Q_1}(E_{i_1, j_1})) | j_3 \rangle. \end{aligned} \quad (240)$$

We now expand the intermediate operator in the matrix-unit basis of $L(\mathcal{H}_{Q_2})$:

$$\begin{aligned} &\Phi_{Q_2|Q_1}(E_{i_1, j_1}) \\ &= \sum_{i_2, j_2} \langle i_2 | \Phi_{Q_2|Q_1}(E_{i_1, j_1}) | j_2 \rangle E_{i_2, j_2}. \end{aligned} \quad (241)$$

Using the linearity of $\Phi_{Q_3|Q_2}$ gives

$$\begin{aligned} & (\varrho_{Q_3|Q_1})^{i_3 i_1}_{j_3 j_1} \\ &= \sum_{i_2, j_2} \langle i_2 | \Phi_{Q_2|Q_1}(E_{i_1, j_1}) | j_2 \rangle \\ & \quad \cdot \langle i_3 | \Phi_{Q_3|Q_2}(E_{i_2, j_2}) | j_3 \rangle. \end{aligned} \quad (242)$$

The two matrix elements are the corresponding components of the two Choi tensors:

$$\langle i_2 | \Phi_{Q_2|Q_1}(E_{i_1, j_1}) | j_2 \rangle = (\varrho_{Q_2|Q_1})^{i_2 i_1}_{j_2 j_1}, \quad (243)$$

$$\langle i_3 | \Phi_{Q_3|Q_2}(E_{i_2, j_2}) | j_3 \rangle = (\varrho_{Q_3|Q_2})^{i_3 i_2}_{j_3 j_2}. \quad (244)$$

It follows that

$$(\varrho_{Q_3|Q_1})^{i_3 i_1}_{j_3 j_1} = \sum_{i_2, j_2} (\varrho_{Q_2|Q_1})^{i_2 i_1}_{j_2 j_1} (\varrho_{Q_3|Q_2})^{i_3 i_2}_{j_3 j_2}. \quad (245)$$

The expression for the action of the composite channel on ρ_{Q_1} was obtained above by substituting Eq. (229) into Eq. (219). $\square$

6. Channel networks with compound quantum registers

The same construction applies when the input and output are compound registers. Let

$$Q = (Q_1, \dots, Q_n), \quad A = (A_1, \dots, A_m), \quad (246)$$

with associated spaces

$$\mathcal{H}_Q = \bigotimes_{r=1}^n \mathcal{H}_{Q_r}, \quad \mathcal{H}_A = \bigotimes_{s=1}^m \mathcal{H}_{A_s}. \quad (247)$$

The corresponding channel network is shown in Fig. 6.

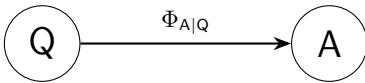


FIG. 6. A channel network with compound input and output registers.

Its Q -factors are

1. $\rho_Q \in D(\mathcal{H}_Q)$;
2. $\rho_{A|Q} \in \text{Pos}(\mathcal{H}_A \otimes \mathcal{H}_Q)$, with

$$\text{Tr}_{\mathcal{H}_A}(\rho_{A|Q}) = \text{Id}_{\mathcal{H}_Q}. \quad (248)$$

For the input register Q , we use the indices $i_1, \dots, i_n$ and $j_1, \dots, j_n$. For the output register A , we use $\alpha_1, \dots, \alpha_m$ and $\beta_1, \dots, \beta_m$. The tensors are

1. $\varrho_Q \in \mathcal{H}_Q \otimes \mathcal{H}_Q^*$, with components $(\varrho_Q)^{i_1 \dots i_n}_{j_1 \dots j_n}$;
2. $\varrho_{A|Q} \in (\mathcal{H}_A \otimes \mathcal{H}_Q) \otimes (\mathcal{H}_A \otimes \mathcal{H}_Q)^*$, with components $(\varrho_{A|Q})^{\alpha_1 \dots \alpha_m i_1 \dots i_n}_{\beta_1 \dots \beta_m j_1 \dots j_n}$;

3. $\varrho_A \in \mathcal{H}_A \otimes \mathcal{H}_A^*$, with components $(\varrho_A)^{\alpha_1 \dots \alpha_m}_{\beta_1 \dots \beta_m}$.

Let $\mathcal{B}_Q$, $\mathcal{B}_A$, and $\mathcal{B}_{A|Q}$ be the canonical bases of $L(\mathcal{H}_Q)$, $L(\mathcal{H}_A)$, and $L(\mathcal{H}_A \otimes \mathcal{H}_Q)$, respectively. We set

$$\begin{aligned} [\varrho_Q]_{\mathcal{B}_Q} &= \rho_Q, & [\varrho_A]_{\mathcal{B}_A} &= \rho_A, \\ [\varrho_{A|Q}]_{\mathcal{B}_{A|Q}} &= \rho_{A|Q}. \end{aligned} \quad (249)$$

The component contraction is

$$\begin{aligned} (\varrho_A)^{\alpha_1 \dots \alpha_m}_{\beta_1 \dots \beta_m} &= (\varrho_{A|Q})^{\alpha_1 \dots \alpha_m i_1 \dots i_n}_{\beta_1 \dots \beta_m j_1 \dots j_n} \\ & \quad \cdot (\varrho_Q)^{i_1 \dots i_n}_{j_1 \dots j_n}. \end{aligned} \quad (250)$$

Equivalently, if

$$[\varsigma_Q]_{\mathcal{B}_Q} = \rho_Q^T, \quad (251)$$

then the same operation takes the upper-lower form

$$\begin{aligned} (\varrho_A)^{\alpha_1 \dots \alpha_m}_{\beta_1 \dots \beta_m} &= (\varrho_{A|Q})^{\alpha_1 \dots \alpha_m i_1 \dots i_n}_{\beta_1 \dots \beta_m j_1 \dots j_n} \\ & \quad \cdot (\varsigma_Q)^{j_1 \dots j_n}_{i_1 \dots i_n}. \end{aligned} \quad (252)$$

C. Classical Bayesian networks in tensor form

We now return to classical variables. As we saw in Section VIA 2, products of classical Choi operators recover the corresponding joint conditional operators once the factors have been placed in a common operator space. Can the same calculation be performed directly with tensors? Since classical probabilistic states and conditional operators are diagonal, their factor product reduces to a pointwise product of their nonzero components. This is the tensor form of the classical factor product [1, Sec. 11.3.1] and coincides with the purely classical instance of the product of Q -factors [3, Sec. 3.1 and Def. 17]. Throughout this subsection we use the explicit-output convention common in programming implementations of Einsum [13, Sec. 2.8]: every index displayed on the left-hand side is a free output index, even if it occurs in several factors on the right-hand side. An index is summed only when it is absent from the output. Thus, indices repeated on both sides are multiplied pointwise rather than contracted.

Consider first the two-node network in Fig. 7.

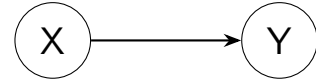


FIG. 7. A classical Bayesian network with two random variables.

For the examples below, we consider binary-valued random variables. We write $\text{Val}(X) = \{x^t, x^f\} \cong \{0, 1\}$, where x^t and x^f denote true and false, respectively, and use the same convention for every other variable. The joint alphabet of Y and X is $\text{Val}(Y) \times \text{Val}(X)$. For each classical node V , we write $\mathcal{H}_V = \mathbb{C}^{\text{Val}(V)}$ for its associated space, as in Sec. IV A 2.

Let $\mathcal{B}_X$ and $\mathcal{B}_{YX}$ denote the canonical bases of $L(\mathcal{H}_X)$ and $L(\mathcal{H}_Y \otimes \mathcal{H}_X)$, respectively. The root factor and the conditional factor are represented by

1. $\varrho_X \in \mathcal{H}_X \otimes \mathcal{H}_X^*$, with components $(\varrho_X)^{xx'}$ and $[\varrho_X]_{\mathcal{B}_X} = \rho_X$;
2. $\varrho_{Y|X} \in (\mathcal{H}_Y \otimes \mathcal{H}_X) \otimes (\mathcal{H}_Y \otimes \mathcal{H}_X)^*$, with components $(\varrho_{Y|X})^{yx'}$ and $[\varrho_{Y|X}]_{\mathcal{B}_{YX}} = \rho_{Y|X}$.

Using this convention, the pointwise product of $\varrho_{Y|X}$ and ϱ_X directly defines $\varrho_{Y,X}$. On the diagonal, this gives

$$(\varrho_{Y,X})^{yx} = (\varrho_{Y|X})^{yx} (\varrho_X)^{xx} \quad (253)$$

Here y and x are output indices, so they are not summed.

The full component expression is

$$(\varrho_{Y,X})^{yx'x'} = (\varrho_{Y|X})^{yx'x'} (\varrho_X)^{xx'}. \quad (254)$$

The factors on the right are diagonal, so all off-diagonal components vanish and the nonzero entries are precisely the joint probabilities. Consequently, $[\varrho_{Y,X}]_{\mathcal{B}_{YX}} = \rho_{Y,X}$.

Example VI.2. Let

$$\begin{aligned} [\varrho_X]_{\mathcal{B}_X} &= \text{diag} \begin{pmatrix} p_X(x^t) \\ p_X(x^f) \end{pmatrix}, \\ [\varrho_{Y|X}]_{\mathcal{B}_{YX}} &= \text{diag} \begin{pmatrix} p_{Y|X}(y^t|x^t) \\ p_{Y|X}(y^t|x^f) \\ p_{Y|X}(y^f|x^t) \\ p_{Y|X}(y^f|x^f) \end{pmatrix}. \end{aligned} \quad (255)$$

Let T be the diagonal type-(2, 2) tensor whose nonzero components are

$$T^{yx}_{yx} = (\varrho_{Y|X})^{yx} (\varrho_X)^{xx}, \quad (256)$$

for $x, y \in \{0, 1\}$, with no summation. Its components are

$$T^{00}_{00} = p_{Y|X}(y^t|x^t) p_X(x^t) = p_{Y,X}(y^t, x^t), \quad (257)$$

$$T^{01}_{01} = p_{Y|X}(y^t|x^f) p_X(x^f) = p_{Y,X}(y^t, x^f), \quad (258)$$

$$T^{10}_{10} = p_{Y|X}(y^f|x^t) p_X(x^t) = p_{Y,X}(y^f, x^t), \quad (259)$$

$$T^{11}_{11} = p_{Y|X}(y^f|x^f) p_X(x^f) = p_{Y,X}(y^f, x^f). \quad (260)$$

Hence $T = \varrho_{Y,X}$.

We now consider the four-node network in Fig. 8.

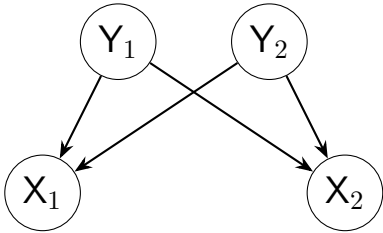


FIG. 8. A classical Bayesian network with four random variables.

The root factors are represented by $\varrho_{Y_r} \in \mathcal{H}_{Y_r} \otimes \mathcal{H}_{Y_r}^*$, with components $(\varrho_{Y_r})^{y_r y'_r}$ for $r \in \{1, 2\}$. For each $r \in \{1, 2\}$, the conditional factor is

$$\varrho_{X_r|Y_1, Y_2} \in (\mathcal{H}_{X_r} \otimes \mathcal{H}_{Y_1} \otimes \mathcal{H}_{Y_2}) \otimes (\mathcal{H}_{X_r} \otimes \mathcal{H}_{Y_1} \otimes \mathcal{H}_{Y_2})^*,$$

with components

$$(\varrho_{X_r|Y_1, Y_2})^{x_r y_1 y_2 x'_r y'_1 y'_2}.$$

Its classical alphabet is $\text{Val}(X_r) \times \text{Val}(Y_1) \times \text{Val}(Y_2)$.

The graph implies $X_1 \perp X_2 \mid (Y_1, Y_2)$, while the two root variables Y_1 and Y_2 are independent. The Bayesian-network factorization therefore gives the following diagonal component identities [1, Sec. 11.2.3, Eq. (11.3)]. Every displayed index is an output index, so no summation is performed. To simplify the notation, let

$$\mathbf{X} = (X_1, X_2), \quad \mathbf{Y} = (Y_1, Y_2), \quad (261)$$

and write

$$\mathbf{x} = (x_1, x_2), \quad \mathbf{y} = (y_1, y_2). \quad (262)$$

The bold indices abbreviate these ordered pairs. Their individual entries remain free, so this notation does not introduce any summation.

$$(\varrho_{\mathbf{X}|\mathbf{Y}})^{\mathbf{xy}}_{\mathbf{xy}} = (\varrho_{X_1|\mathbf{Y}})^{x_1 \mathbf{y}}_{x_1 \mathbf{y}} (\varrho_{X_2|\mathbf{Y}})^{x_2 \mathbf{y}}_{x_2 \mathbf{y}}. \quad (263)$$

$$\begin{aligned} (\varrho_{\mathbf{X}, Y_1 | Y_2})^{\mathbf{xy}}_{\mathbf{xy}} &= (\varrho_{X_1|\mathbf{Y}})^{x_1 \mathbf{y}}_{x_1 \mathbf{y}} (\varrho_{X_2|\mathbf{Y}})^{x_2 \mathbf{y}}_{x_2 \mathbf{y}} \\ &\quad \cdot (\varrho_{Y_1})^{y_1}_{y_1}. \end{aligned} \quad (264)$$

$$\begin{aligned} (\varrho_{\mathbf{X}, \mathbf{Y}})^{\mathbf{xy}}_{\mathbf{xy}} &= (\varrho_{X_1|\mathbf{Y}})^{x_1 \mathbf{y}}_{x_1 \mathbf{y}} (\varrho_{X_2|\mathbf{Y}})^{x_2 \mathbf{y}}_{x_2 \mathbf{y}} \\ &\quad \cdot (\varrho_{Y_1})^{y_1}_{y_1} (\varrho_{Y_2})^{y_2}_{y_2}. \end{aligned} \quad (265)$$

These identities illustrate one of the main advantages of the tensor formulation. Unlike the matrix formalism in Section VI A 2, there is no need to construct the full lifted matrices before multiplying the factors. The indices determine the variables shared by the factors, and a factor that does not contain one of the output indices is simply multiplied across all values of that index. The joint probabilities are therefore obtained directly from the pointwise product of the corresponding components.

D. Hybrid classical–quantum networks

1. Hybrid operators

We now consider networks containing both classical and quantum nodes. Their local factors are operators on tensor products of classical and quantum register spaces. We place all classical registers before all quantum registers and use the same order for their component indices. When this differs from the output–input order of Sec. V C 1, we use the canonical permutation of the tensor factors. All transposes are taken in the chosen orthonormal bases.

Let $\mathbf{X}$ be a classical node with alphabet Σ . Its associated space is $\mathcal{H}_X = \mathbb{C}^\Sigma$, with standard basis $\{|x\rangle : x \in \Sigma\}$ indexed by the possible values of $\mathbf{X}$. Let $\mathbf{Q}$ be a quantum node with associated space $\mathcal{H}_Q$. The composite space is

$$\mathcal{H}_{XQ} = \mathcal{H}_X \otimes \mathcal{H}_Q. \quad (266)$$

Definition VI.1. A *hybrid operator* on $\mathcal{H}_{XQ}$ is an operator of the form

$$\sigma_{XQ} = \sum_{x \in \Sigma} |x\rangle\langle x|_X \otimes M_x^Q, \quad (267)$$

where $\{M_x^Q : x \in \Sigma\}$ is a family of operators on $\mathcal{H}_Q$, indexed by the values of X [2, Def. III.10].

When X is a parent of Q , the corresponding conditional factor has the form

$$\sigma_{Q|X} = \sum_{x \in \Sigma} |x\rangle\langle x|_X \otimes \rho_x^Q, \quad (268)$$

where $\rho_x^Q \in D(\mathcal{H}_Q)$ for every $x \in \Sigma$. It is positive and conditionally normalized:

$$\text{Tr}_{\mathcal{H}_Q}(\sigma_{Q|X}) = \text{Id}_{\mathcal{H}_X}. \quad (269)$$

When Q is a parent of X , a POVM $\{P_x^Q : x \in \Sigma\}$ defines the conditional factor

$$\sigma_{X|Q} = \sum_{x \in \Sigma} |x\rangle\langle x|_X \otimes (P_x^Q)^T, \quad (270)$$

which is positive and satisfies

$$\text{Tr}_{\mathcal{H}_X}(\sigma_{X|Q}) = \text{Id}_{\mathcal{H}_Q}. \quad (271)$$

When the components M_x^Q are positive, this block-diagonal operator encodes the Q -factor $x \mapsto M_x^Q$ [3, Def. 13]. The transpose in $\sigma_{X|Q}$ follows from the Choi convention of Sec. VC1: the measurement channel $A \mapsto \sum_x \text{Tr}(P_x^Q A) |x\rangle\langle x|_X$ has this Choi operator.

2. A preparation–measurement network

Consider the hybrid network

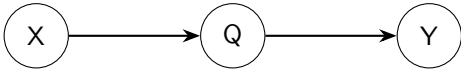


FIG. 9. A preparation–measurement network with classical nodes X and Y and quantum node Q .

Let Σ and Γ be the alphabets of X and Y , respectively. The associated Q -factors are

1. the classical factor associated with the node X is ρ_X ;
2. the hybrid factor associated with the node Q is $\rho_{Q|X}$;
3. the hybrid factor associated with the node Y is $\rho_{Y|Q}$.

Here, p_X is the distribution encoded by ρ_X , the states $\{\rho_x^Q : x \in \Sigma\}$ define the preparation factor, and the POVM $\{P_y^Q : y \in \Gamma\}$ defines the measurement factor as in Definition VI.1.

We now use the operator-to-tensor representation described in Sec. IIIB. Choose a fixed orthonormal basis $\{|q\rangle : q \in \Lambda\}$ of $\mathcal{H}_Q$. Let $\mathcal{B}_X$, $\mathcal{B}_{XQ}$, and $\mathcal{B}_{YQ}$ denote the canonical bases of $L(\mathcal{H}_X)$, $L(\mathcal{H}_X \otimes \mathcal{H}_Q)$, and $L(\mathcal{H}_Y \otimes \mathcal{H}_Q)$, respectively. Quantum indices are contracted according to the coordinate convention of Sec. VIB5. For classical indices, we use the explicit-output convention of Sec. VIC, illustrated in Eq. 253: indices displayed on the left-hand side remain free and are multiplied pointwise. The tensors corresponding to the three Q -factors are

1. $\varrho_X \in \mathcal{H}_X \otimes \mathcal{H}_X^*$, with components $(\varrho_X)^{x_{x'}}$ and

$$[\varrho_X]_{\mathcal{B}_X} = \rho_X; \quad (272)$$

2. $\varrho_{Q|X} \in (\mathcal{H}_X \otimes \mathcal{H}_Q) \otimes (\mathcal{H}_X \otimes \mathcal{H}_Q)^*$, with components $(\varrho_{Q|X})^{xq_{x'q'}}$ and

$$[\varrho_{Q|X}]_{\mathcal{B}_{XQ}} = \rho_{Q|X}; \quad (273)$$

3. $\varrho_{Y|Q} \in (\mathcal{H}_Y \otimes \mathcal{H}_Q) \otimes (\mathcal{H}_Y \otimes \mathcal{H}_Q)^*$, with components $(\varrho_{Y|Q})^{yq_{y'q'}}$ and

$$[\varrho_{Y|Q}]_{\mathcal{B}_{YQ}} = \rho_{Y|Q}. \quad (274)$$

Combining the three Q -factors using this component convention gives a tensor on the two classical nodes X and Y :

$$(\varrho_{X,Y})^{xy_{x'y'}} = (\varrho_X)^{x_{x'}} (\varrho_{Q|X})^{xq_{x'q'}} (\varrho_{Y|Q})^{yq_{y'q'}}. \quad (275)$$

Because the measurement is modeled as destructive, $\varrho_{Y|Q}$ has quantum input indices but no quantum output indices. Its indices q and q' are contracted with those introduced by $\varrho_{Q|X}$. On the diagonal,

$$\begin{aligned} (\varrho_{X,Y})^{xy_{xy}} &= p_X(x) \sum_{q,q'} (\rho_x^Q)^q_{q'} \left((P_y^Q)^T \right)^q_{q'} \\ &= p_X(x) \text{Tr}(\rho_x^Q \rho_y^Q). \end{aligned} \quad (276)$$

All off-diagonal classical components vanish. Since $\sum_y P_y^Q = \text{Id}_{\mathcal{H}_Q}$, these components form a normalized joint distribution on (X, Y) .

VII. CONCLUSION AND OUTLOOK

We began with the question of whether the local factors associated with classical and quantum nodes could be described within the same operator formalism. Encoding finite probability distributions as diagonal density operators provides such a description. A conditional distribution then determines a positive operator that is exactly the Choi operator of the quantum channel induced by the corresponding stochastic map. Classical conditional factors therefore appear as a special case of the channel factors used for quantum nodes.

In fixed bases, the tensor representation makes the action and composition of these factors more direct. The

transpose, operator multiplication, and partial trace in the usual Choi action are collected into a contraction over the input indices, while the intermediate register remains explicit in a composition of channels. For diagonal operators, multiplication reduces to the pointwise product of probability factors. In the hybrid example, the contraction between a preparation and a destructive measurement removes the quantum indices and leaves a factor on the classical registers.

We have focused on the representation and combination of local factors, rather than on a complete theory of inference for quantum Bayesian networks. The component formulation depends on the chosen orthonormal bases and on a fixed ordering of the registers. A natural continuation would be to extend these rules to general hybrid directed acyclic graphs and study how they interact with marginalization, conditioning, and inference.

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- [1] A. Darwiche, in *Handbook of Knowledge Representation*, Foundations of Artificial Intelligence, Vol. 3, edited by F. van Harmelen, V. Lifschitz, and B. Porter (Elsevier, 2008) pp. 467–509.
 - [2] M. S. Leifer and R. W. Spekkens, *Physical Review A* **88**, 052130 (2013), [arXiv:1107.5849 \[quant-ph\]](#).
 - [3] R. Di Guardia, T. Ehrhard, and C. Faggian, in *11th International Conference on Formal Structures for Computation and Deduction (FSCD 2026)*, Leibniz International Proceedings in Informatics (LIPIcs), Vol. 378, edited by F. Pfenning (Schloss Dagstuhl – Leibniz-Zentrum für Informatik, Dagstuhl, Germany, 2026) pp. 16:1–16:23.
 - [4] J. Watrous, *The Theory of Quantum Information* (Cambridge University Press, 2018).
 - [5] W. Greub, *Multilinear Algebra*, 2nd ed., Universitext (Springer, New York, 1978).
 - [6] C. Wendl, *Multilinear algebra and index notation*, Appendix A to *Lecture Notes on Bundles and Connections* (2008), last updated September 26, 2008.
 - [7] K. Dullemond and K. Peeters, *Introduction to tensor calculus* (2010), english translation, re-typeset in 2008/2010.
 - [8] B. Conrad, *Isomorphisms and tensor products*, Math 396 course handout, Stanford University (2006).
 - [9] M. Leifer, The Choi–Jamiołkowski isomorphism: You’re doing it wrong!, [mattleifer.info](#) (2011), blog post.
 - [10] A. Jamiołkowski, *Reports on Mathematical Physics* **3**, 275 (1972).
 - [11] M.-D. Choi, *Linear Algebra and its Applications* **10**, 285 (1975).
 - [12] G. Chiribella, G. M. D’Ariano, and P. Perinotti, *Physical Review A* **80**, 022339 (2009), [arXiv:0904.4483 \[quant-ph\]](#).
 - [13] T. Rocktäschel, Einsum is all you need: Einstein summation in deep learning, <https://rockt.ai/2018/04/30/einsum> (2018), blog post.